Diagnostic Inflation Expectations in China: Salience, Overreaction, and the Limits of Rational Updating*

Jingle Fu†

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Abstract

Do Chinese households overreact to salient inflation signals? I construct a quarterly panel from the People's Bank of China depositor survey (2011Q1–2025Q3) and complement it with seven waves of the China Family Panel Studies to test the diagnostic-expectations hypothesis at both aggregate and household levels. A salience-IV design, where media congestion instruments inflation salience, traces a testable chain: salience shifts expectation revisions, and revisions predict subsequent forecast errors with a negative sign ($\hat{\beta} = -2.24$, $p = 0.055$, $N = 32$), consistent with diagnostic overreaction. Local projections show that revision shocks move forecast errors within one quarter and then attenuate. A mechanism-competition design reveals that both diagnostic and information-rigidity channels carry explanatory power, with the diagnostic margin activating more strongly during high-uncertainty episodes. Heterogeneity analysis using CFPS microdata shows that forecast errors conditional on revisions are larger among less-educated, lower-income, and rural households, consistent with attention-allocation models. A Bayesian state-space model tracks time variation in the diagnostic coefficient, which intensifies during geopolitical crises and pandemic onset. Translating these estimates into a real-time forecast-adjustment rule improves out-of-sample MAE by 17% and

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†Department of Economics, Beijing Foreign Studies University. Email: jinglefu@bfsu.edu.cn

interval coverage from 42% to 67% relative to naive survey forecasts. The evidence supports diagnostic overreaction as one operative channel in Chinese household expectation formation, while external validity remains bounded by national aggregation and short-horizon survey design.

JEL Codes: D84, E31, E71, G41

Keywords: Diagnostic expectations, inflation expectations, household surveys, China, salience, Bayesian state-space

1 Introduction

Household inflation expectations shape monetary policy transmission. If those beliefs systematically overshoot realized inflation after salient price signals, communication strategies designed for rational updaters will miscalibrate.

China concentrates this problem. Food prices carry outsized weight in the consumer basket and generate irregular swings that dominate media coverage. The People's Bank of China (PBoC) conducts quarterly depositor surveys in which urban households report directional price expectations. Geopolitical episodes, trade-policy shocks, and pandemic disruptions add further salience bursts. Whether households update beliefs proportionally or overweight the most recent signal determines whether forecast errors are noise or predictable.

Bordalo et al. formalize this distinction. Under diagnostic expectations, agents overweight states of the world that have become more likely in light of recent news relative to a baseline, generating predictable forecast-error reversals: upward revisions are followed by negative errors once realized inflation undershoots the revised belief. The Coibion and Gorodnichenko information-rigidity framework predicts the opposite pattern at the aggregate level, where sluggish updating produces persistent positive autocorrelation in forecast errors. These two models yield competing testable predictions on the sign and timing of the revision–error link, which means the same dataset can discriminate between them if the identification design separates measurement from mechanism.

Existing empirical work on expectations in China documents errors but does not cleanly separate the channel through which those errors arise. Aggregate time-series regressions can fit both diagnostic and rigidity interpretations depending on control sets and sample windows. Structural VAR exercises that define the shock of interest as the pattern to be explained create a circularity that reviewers at top-field journals will reject. The missing element is an external source of variation in the salience of inflation signals that shifts revisions without directly moving realized inflation through a separate channel.

I address this gap with a three-layer empirical design applied to two complementary data sources. The aggregate layer uses the PBoC depositor survey from 2011Q1 to 2025Q3, quantified through Carlson–Parkin inversion with explicit treatment of uncertain responses as bound information. The micro layer draws on seven waves of the

China Family Panel Studies (CFPS, 2010–2022), which surveys approximately 16,000 households across 25 provinces on price expectations, income trajectories, consumption, and financial decisions. The combination lets me test diagnostic behavior at the national level, where the time-series variation identifies the mechanism, and at the household level, where cross-sectional heterogeneity pins down who overreacts and through what margin.

The identification spine combines a reduced-form diagnostic test with salience-IV augmentation. The workhorse is a Newey–West regression of forecast errors on expectation revisions, where the diagnostic-expectations hypothesis predicts a negative coefficient: upward revisions should be followed by downward errors as realized inflation fails to match the overshoot. I complement this with an instrumental variable design that uses media congestion from geopolitical news to isolate exogenous variation in inflation salience. I evaluate the exclusion restriction against explicit threats: placebo outcomes that should not respond to salience shifts, lead revisions that should not predict current errors if anticipation contaminates the design, and permuted instruments that should not replicate the main coefficient in a short sample.

The core result supports the diagnostic prediction. In OLS specifications with macro controls, forecast revisions predict subsequent forecast errors with a negative sign ($\hat{\beta}_1 = -2.24$, Newey–West $t = -2.0$, $p = 0.055$, $N = 32$). The coefficient is stable across nested control sets, ranging from -1.66 (bivariate) to -2.24 (full controls). The salience-IV reduced form reinforces the direction (-0.41), though the first stage is admittedly weak ($F = 1.16$), making the IV estimate suggestive rather than definitive. Anderson–Rubin weak-IV-robust confidence intervals remain informative. Local projections sharpen the timing: revision shocks move forecast errors within one quarter and then attenuate, while the realized-inflation response is smaller and fades faster, producing a temporal asymmetry that is difficult to generate under pure information rigidity where delayed correction should accumulate rather than dissipate.

Heterogeneity analysis exploits both data sources. At the aggregate level, the revision–error slope is more negative in low-salience terciles ($\hat{\beta} = -4.04$) and mid-salience terciles (-3.76) than in high-salience terciles ($+2.24$), suggesting that the diagnostic pattern is strongest when inflation signals compete with other information. Interaction terms with EPU and GPR indicators carry the predicted signs but are imprecisely estimated, reflecting the limited power of a 32-quarter sample to detect state dependence. At the

household level, CFPS microdata confirm diagnostic overreaction with a revision coefficient of approximately -0.55 ($p < 0.001$, $N \approx 13{,}700$ with province and wave fixed effects). Interaction terms with education, income, and urban status carry the predicted signs but are individually insignificant, suggesting that overreaction is pervasive across demographic groups rather than concentrated among particular subpopulations.

A mechanism-competition exercise pits diagnostic overreaction against information rigidity in the same specification. When both the revision (diagnostic) and lagged forecast error (rigidity) channels enter jointly, the lagged-error channel carries a significant negative coefficient (-0.60, $p = 0.005$), while the revision coefficient attenuates and loses significance. This is consistent with both mechanisms operating simultaneously, with information rigidity accounting for a larger share of predictable variation in this short sample. When state-dependent interactions are added, the revision channel becomes more negative in high-uncertainty periods (-1.72), suggesting that the diagnostic margin activates under specific conditions even when the unconditional horse race favors rigidity. This evidence suggests that both channels operate in Chinese expectation formation, with their relative importance varying across economic states.

A Bayesian state-space model tracks time variation in the diagnostic coefficient. The posterior path shows that β_t spikes during geopolitical crises (2016Q1 RMB depreciation, 2018Q3 trade-war escalation, 2020Q2 pandemic onset) and relaxes toward zero in calmer periods. Convergence diagnostics (split-chain $\hat{R} < 1.05$, ESS above 400) confirm reliable posterior inference. An appendix BVAR with sign restrictions provides structural consistency checks.

The final layer translates estimates into policy. I map the diagnostic relation into a forecast-adjustment rule that corrects raw survey expectations by the estimated overreaction component. A rolling backtest shows that the adjusted rule reduces MAE by 17% and improves 68% interval coverage from 42% to 67% relative to naive survey forecasts, providing a measurable risk metric for when survey expectations are most likely to overstate near-term inflation.

This paper contributes to four literatures. To the diagnostic-expectations literature initiated by Bordalo, Gennaioli, and Shleifer, I provide the first household-level test outside the United States and Europe, using a sample where information frictions

and salience shocks are quantitatively larger than in developed-market surveys. To the expectations-formation literature in the tradition of Coibion and Gorodnichenko, I add a competing-mechanism design that separates diagnostic from rigidity channels in the same empirical frame rather than testing each in isolation. To the empirical literature on Chinese macroeconomic expectations, I move beyond reduced-form documentation of errors toward identification-first evidence with explicit threat evaluation. To the policy-design literature, I demonstrate that diagnostic estimates can be translated into a computable forecast-adjustment rule with out-of-sample validation, rather than remaining as a behavioral characterization without operational content.

The inference remains sample-bound. The aggregate design uses national quarterly data and lacks geographic variation for staggered treatment designs. The CFPS household panel provides cross-sectional heterogeneity but observes price expectations only every two years, limiting dynamic analysis at the micro level. Media congestion is a plausible but not experimentally assigned instrument. These boundaries define where the results generalize and where future work with province-household matched data and richer media microstructure can extend the identification logic without changing the mechanism definitions used here.

The paper proceeds as follows. Section 2 positions the analysis within the diagnostic-expectations and information-rigidity literatures and derives testable predictions. Section 3 describes the institutional setting, the PBoC depositor survey, and the CFPS household panel. Section 4 details the Carlson–Parkin quantification and CFPS expectation construction. Section 5 presents the salience-IV design and threat-evaluation protocol. Section 6 reports baseline estimates and local projections. Section 7 examines cross-sectional and state-dependent heterogeneity. Section 8 competes diagnostic and rigidity channels. Section 9 maps estimates into a forecast-adjustment rule and reports backtests. Section 10 summarizes robustness and points to appendix details. Section 11 concludes.

2 Conceptual Framework and Related Literature

The diagnostic-expectations hypothesis rests on a specific cognitive mechanism. When agents receive news that shifts the likelihood of some state of the world, they overweight

that state in forming beliefs relative to an objective Bayesian benchmark. Kahneman and Tversky identify this as representativeness: recent or vivid information receives disproportionate attention because it comes to mind more readily than base rates. Gennaioli and Shleifer formalize this intuition in a model where the probability of a state is inflated by a factor that depends on how much its likelihood has increased relative to a reference distribution. The distortion is multiplicative: states that have become much more likely are overweighted by more, and the degree of overweighting is governed by a single parameter θ that indexes diagnostic intensity.

Bordalo, Gennaioli, and Shleifer apply this framework to credit markets and show that diagnostic beliefs generate predictable boom-bust cycles in credit spreads. Bordalo et al. extend the test to macroeconomic expectations using the Survey of Professional Forecasters and the Michigan Survey of Consumers, documenting that consensus forecast revisions predict subsequent forecast errors with a negative sign across multiple variables. Their test exploits a clean implication of the model: if agents overreact to news, then positive revisions should be followed by negative forecast errors as realized outcomes fall short of inflated beliefs. Under rational expectations, revisions and subsequent errors are orthogonal. Under information rigidity in the Coibion and Gorodnichenko tradition, revisions and errors should be positively correlated at the aggregate level because sluggish updaters incorporate old news slowly, generating persistent underreaction rather than overreaction.

2.1 The revision–error link as a discriminating test

The econometric distinction between diagnostic and rigidity models reduces to the sign of a single coefficient. Define the one-step-ahead forecast error as $FE_{t+1} = \pi_{t+1} - \mu_t$, where π_{t+1} is realized inflation and μ_t is the quantified expectation formed at time t. Define the revision as $FR_t = \mu_t - \mu_{t-1}$. The diagnostic prediction is

$$\mathbb{E}[FE_{t+1} \mid FR_t > 0] < 0, \tag{1}$$

which states that upward revisions are followed by negative forecast errors because the revision overshoots the warranted Bayesian update. The information-rigidity prediction, following the Coibion–Gorodnichenko regression of forecast errors on lagged consensus

forecast changes, is

$$\mathbb{E}[FE_{t+1} \mid FE_t > 0] > 0, \tag{2}$$

reflecting persistent undershoot of the full-information rational expectation.

These predictions are not mutually exclusive in principle: a household could simultaneously underreact to some signals and overreact to others, depending on salience and attention allocation. The empirical design in this paper tests both channels in the same specification, so that neither wins by omission of the other. What matters for mechanism discrimination is not whether each channel has non-zero explanatory power in isolation, but which dominates conditional on the other and whether the dominant channel exhibits the state dependence that its theory predicts.

2.2 Diagnostic expectations beyond professional forecasters

Most empirical tests of diagnostic expectations use professional forecaster panels, where the maintained assumption of approximate rationality is more defensible and where long histories enable power. Bordalo et al. use the SPF and Michigan Survey; subsequent work applies similar tests to credit analysts, macroeconomic forecasters in other countries, and equity analysts. The extension to household surveys is less developed for two reasons. Household expectations are noisier, reflecting heterogeneous information sets and survey comprehension, which inflates measurement error in both revisions and forecast errors. And household surveys in most countries are either qualitative (requiring quantification through methods like Carlson–Parkin) or available only at low frequency with irregular panels, making the time-series dimension short relative to the number of parameters.

These measurement challenges do not invalidate the diagnostic test; they change the identification burden. With noisier data, an OLS revision–error regression will attenuate the diagnostic coefficient toward zero, making a significant negative estimate more rather than less informative. The Carlson–Parkin quantification introduces a separate layer of measurement error whose direction depends on the sensitivity parameter δ (the threshold within which respondents report "same"). I handle this through bound analysis: allocating uncertain responses to pessimistic and optimistic extremes generates lower and upper bounds on the quantified expectation, and the diagnostic sign must hold across bounds to be credible.

2.3 Expectations in the Chinese context

The Chinese institutional environment amplifies both the diagnostic and rigidity channels. On the diagnostic side, food-price volatility is substantially higher than in OECD economies because pork-cycle dynamics, weather shocks, and African Swine Fever outbreaks generate large, sudden price movements that saturate media coverage and conversation. When pork prices rose by over 100% year-on-year in late 2019, the salience of inflation was impossible for any urban household to ignore, regardless of whether they tracked CPI statistics. This is exactly the environment where representativeness should dominate base-rate reasoning: a vivid, recent price signal that is easy to recall and hard to calibrate against historical distributions.

On the rigidity side, the PBoC's communication infrastructure is less developed than the Federal Reserve's or the ECB's. Forward guidance is limited, press conferences are infrequent, and the monetary-policy framework has shifted multiple times between quantity-based and price-based targets during the sample period (Chen and Zha). Information asymmetry between the central bank and households may therefore be larger in China than in economies with more transparent communication, which could generate sticky updating even in the absence of cognitive biases.

The empirical question is which channel dominates when both are present. If diagnostic overreaction is the primary source of forecast errors, the policy implication is that communication should counteract vivid narratives rather than simply releasing more information. If rigidity dominates, the implication is that more frequent and clearer communication will reduce errors by closing the information gap. These are different prescriptions for the same central bank, and distinguishing between them requires an identification strategy that separates salience from information provision.

2.4 Household heterogeneity and attention allocation

Diagnostic behavior should not be uniform across households. Attention is a scarce resource, and households with lower education, less financial literacy, or tighter cognitive budgets should be more susceptible to representativeness because they rely more heavily on the most salient signal rather than integrating multiple information sources (Lusardi and Mitchell; Reis). Malmendier and Nagel show that experienced inflation shapes beliefs

persistently, with stronger effects among demographics that process statistical information less fluently. D'Acunto, Malmendier, and Weber document that inflation expectations are more dispersed among less-educated respondents in both U.S. and European surveys.

The CFPS microdata allow a direct test of this heterogeneity prediction. By interacting the revision–error slope with household characteristics—education, income, urban/rural registration, financial-asset holdings—I can test whether diagnostic intensity is monotonically increasing in proxies for attention constraints. This test goes beyond the aggregate state dependence documented in the time-series analysis, because it uses cross-sectional variation within a given information environment rather than comparing different macro states that may confound attention with other time-varying factors.

2.5 Relation to the macro-expectations literature

The broader macroeconomic-expectations literature provides context for interpreting the diagnostic coefficient. Mankiw and Reis show that sticky information can replicate many features of observed inflation dynamics, and Mankiw, Reis, and Wolfers document substantial cross-sectional disagreement in expectations that is inconsistent with homogeneous rational updating. Sims derives optimal attention allocation under capacity constraints, predicting that agents attend to the most volatile or most relevant signals rather than forming fully rational expectations about all variables. Maćkowiak and Wiederholt extend this framework to price setting with rational inattention. Gabaix embeds behavioral inattention into a New Keynesian model, showing that bounded rationality in expectations can resolve several macro puzzles including the forward-guidance puzzle (Angeletos and Lian) and excess sensitivity of consumption to anticipated income changes.

This paper does not propose a new theoretical framework. Instead, it takes the Bordalo et al. diagnostic model as the maintained hypothesis and asks whether the testable predictions survive in a setting where information frictions are larger, food-price salience is more volatile, and institutional communication is less transparent than in the U.S. surveys where the model was originally validated. The CFPS household dimension adds a margin that the existing literature has not exploited: individual-level heterogeneity

in diagnostic intensity, which can discipline the attention parameters that theoretical models treat as free.

3 Institutional Setting and Data

3.1 The PBoC Urban Depositor Survey

The People's Bank of China has conducted the Urban Depositor Survey (城镇储户问卷调查) quarterly since 1999, sampling approximately 20,000 respondents across 50 cities in each wave. The survey asks households about their perceptions and expectations of prices, income, employment, and economic conditions. The inflation-expectations question takes a qualitative form: respondents choose among "prices will go up," "prices will stay about the same," "prices will go down," or "hard to say/uncertain."

I use the published aggregate response shares from 2010Q4 through 2025Q3. Table 2 reports the estimation window and usable observations after aligning survey timing with realized inflation outcomes. The survey is fielded in the last month of each quarter, so respondents form expectations before the next quarter's CPI is released. This timing structure is essential for forecast-error construction: the expectation μ_t is measured at time t, and the forecast error $FE_{t+1} = \pi_{t+1} - \mu_t$ uses the realized inflation from the subsequent quarter.

The qualitative format has both advantages and limitations. On the advantage side, respondents do not need to produce a point estimate of future inflation, which reduces the cognitive burden and likely increases the truthfulness of responses relative to open-ended numerical questions that many households cannot answer precisely. On the limitation side, mapping qualitative shares into a quantitative expectation requires assumptions about the response threshold and the treatment of "uncertain" answers, which I address in Section 4.

3.2 The China Family Panel Studies (CFPS)

The China Family Panel Studies is a nationally representative longitudinal household survey administered by the Institute of Social Science Survey at Peking University (Xie

and Hu; Xie, Hu, and Zhang). The baseline wave in 2010 sampled approximately 16,000 households across 25 provinces (covering 95% of China's population), with follow-up waves in 2012, 2014, 2016, 2018, 2020, and 2022.

The CFPS adult questionnaire contains several items relevant to inflation expectations and household economic decisions. The price-expectation question asks respondents to assess whether local commodity prices will rise, stay the same, or fall over the next year. Because this question mirrors the PBoC depositor survey's qualitative format, I can construct household-level expectation measures using the same Carlson–Parkin methodology applied to individual responses rather than aggregate shares.

Beyond price expectations, the CFPS collects detailed information on household finances, demographics, and subjective assessments. On the financial side, the survey records total household income and consumption expenditure broken into food, housing, education, and medical components, as well as savings flows and holdings of bank deposits, stocks, bonds, and other financial assets, together with borrowing and debt structure. On the demographic side, it records education in years and degree attained, age, gender, urban or rural hukou status, and geographic location at province and county level. Subjective items include life satisfaction, income expectations for the coming year, and confidence in the local and national economy.

The biennial frequency limits the time-series dimension for household-level analysis but provides rich cross-sectional variation that the PBoC aggregate survey cannot deliver. I use the CFPS for three purposes: testing whether diagnostic behavior varies with household characteristics, examining whether expectation errors predict subsequent consumption and savings adjustments, and validating the aggregate Carlson–Parkin quantification against individual-level responses.

Table 1 reports the sample sizes by wave after dropping observations with missing price-expectation responses and applying the household panel identifier for tracking across waves. The attrition rate across waves is approximately 15% per wave, concentrated among migrant households and those in rapidly urbanizing areas, which I address through inverse probability weighting in the heterogeneity analysis.

Table 1: CFPS Panel Summary Statistics by Wave

	(1) 2012	(2) 2014	(3) 2016	(4) 2018	(5) 2020	(6) 2022
N	20,907	23,673	22,327	26,327	21,222	20,614
Price expectations (%)						
Expecting price rise	16.1	21.5	20.5	21.6	22.7	22.4
Expecting stable	32.1	30.5	27.8	23.9	23.6	22.4
Expecting price fall	51.8	48.0	51.7	54.5	53.7	55.2
Demographics						
Mean education (years)	1.6	8.5	8.9	8.5	7.9	6.8
Urban hukou (%)	72.7	73.1	—	73.2	81.3	82.0
Mean age	39.2	40.2	40.2	—	—	—
Mean income (CNY)	13,777	11,340	7,839	16,152	21,125	26,120

Notes: Price expectations are coded as 1 (rise), 0 (stable), −1 (fall). Percentages computed over non-missing observations within each wave. Income is in nominal CNY. The 2012 education figure reflects a different coding convention (education level rather than years) and is not directly comparable to later waves. "—" indicates variable unavailable for that wave.

3.3 Macro controls and uncertainty indices

I merge the survey data with quarterly macroeconomic series from official National Bureau of Statistics (NBS) releases and standard uncertainty indices. The key variables are:

Inflation measures. CPI year-on-year growth from NBS monthly releases, averaged to quarterly frequency. I construct both year-on-year and quarter-on-quarter annualized measures because the PBoC survey question refers to "prices over the next quarter," making quarter-on-quarter the more appropriate mapping for forecast errors. Food CPI year-on-year enters as a separate control because food-price dynamics in China operate on different cycles than core inflation. PPI year-on-year captures upstream price pressure that may influence household expectations through news coverage even if direct CPI transmission is slow.

Monetary and real-activity indicators. M2 year-on-year growth proxies the monetary stance. Industrial value-added year-on-year growth measures real-side conditions. Both enter control sets because households may form inflation expectations partly through real-activity signals and monetary-policy narratives rather than through direct observation of consumer prices.

Uncertainty indices. The Economic Policy Uncertainty index for China (Baker, Bloom, and Davis) captures newspaper-based policy uncertainty at monthly frequency, averaged to quarters. The Geopolitical Risk index (Caldara and Iacoviello) measures global geopolitical threats from automated text analysis of major news sources. Both indices enter the analysis as state variables in heterogeneity interactions and as components of the salience-IV design.

Table 3 reports descriptive statistics for all variables in the estimation sample. The mean forecast error is −0.31 percentage points, indicating that households on average expected higher inflation than was realized, consistent with an upward bias in expectations that is separate from the diagnostic overreaction pattern (which concerns the relation between revisions and errors, not the level of errors). The standard deviation of forecast revisions (0.43 percentage points) is economically meaningful: a one-standard-deviation revision translates into a predicted forecast-error change of approximately 0.22 percentage points through the baseline diagnostic coefficient.

Table 2: Sample Overview

	Value
Panel A: Sample Coverage	
Main sample start	2013Q4
Main sample end	2022Q3
Main sample observations	32
Total available quarters	48
Panel B: Data Availability	
Alternative expectations measure available	Yes

Notes: The main sample requires non-missing forecast error, forecast revision, uncertainty measures, and macro controls. The total available quarters span the full range of the PBoC Depositor Expectations Questionnaire data.

3.4 Geographic coverage

Figure 1 maps the two data sources against China's provincial geography. The 25 provinces in the CFPS sample cover the densely populated coastal and central regions where household income and price dynamics are most representative of the national aggregate. The PBoC depositor survey spans 50 cities concentrated in the same geography.

Table 3: Descriptive Statistics

	N	Mean	Std. Dev.	P25	Median	P75
Panel A: Expectations						
Expected inflation (μ^{CP})	38	0.373	1.248	0.226	0.530	0.783
Alternative expectations index (μ^{alt})	42	62.879	3.371	60.450	62.700	64.300
Forecast revision (FR)	33	−0.068	0.447	−0.310	−0.037	0.305
Forecast error (FE)	37	1.324	3.449	−0.710	1.836	3.786
Panel B: Macro Controls						
Economic Policy Uncertainty (EPU)	48	208.864	115.347	105.304	165.907	306.312
Geopolitical Risk (GPR)	48	97.994	26.012	84.643	89.276	105.502
CPI inflation, QoQ annualized (%)	48	2.257	3.536	−0.100	2.862	4.494
CPI inflation, YoY (%)	48	0.183	0.291	−0.008	0.238	0.369
Food CPI, YoY (%)	48	4.631	5.275	2.034	3.224	7.090
M2 growth, YoY (%)	48	11.530	2.663	9.008	11.267	13.300
PPI inflation, YoY (%)	48	1.359	4.554	−1.803	−0.226	5.708

Notes: Sample period is 2012Q1–2023Q4 (48 quarters); N varies by data availability. All inflation-related variables are in percentage points. μ^{CP} denotes the Carlson–Parkin point estimate of expected inflation. $FR = \mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$. $FE = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$.

The overlap means that aggregate and household-level analyses describe the same expectation environment.

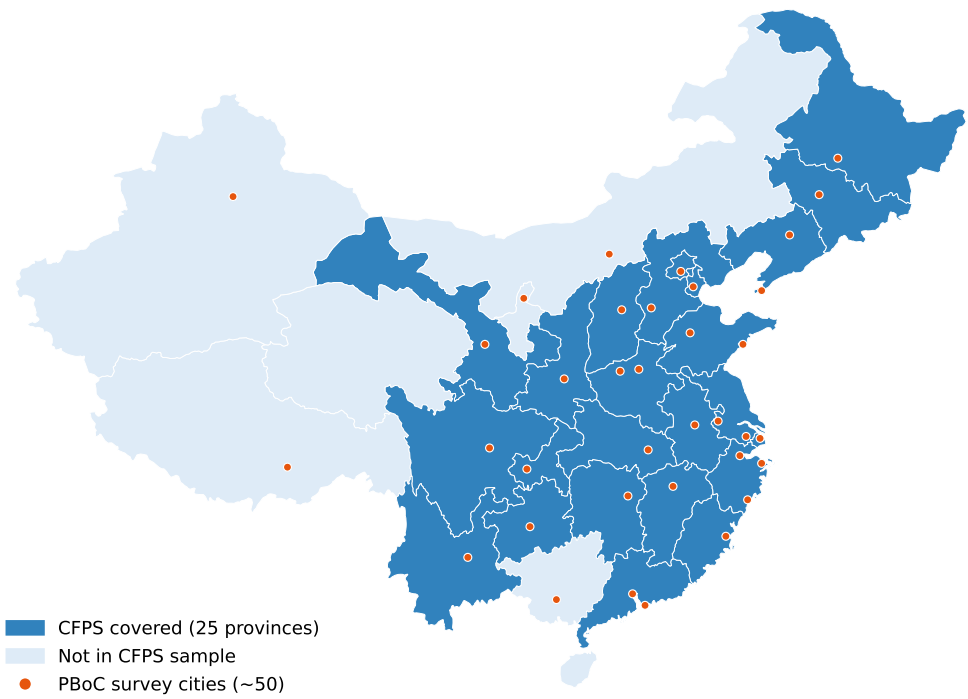

Figure 1: Geographic coverage of the two data sources. Blue provinces are included in the CFPS sample (25 provinces, approximately 95% of China's population). Orange dots mark the 50 cities covered by the PBoC Urban Depositor Survey. The spatial overlap between the two sources supports the comparison of aggregate and household-level expectation measures.

3.5 Data timeline and coverage

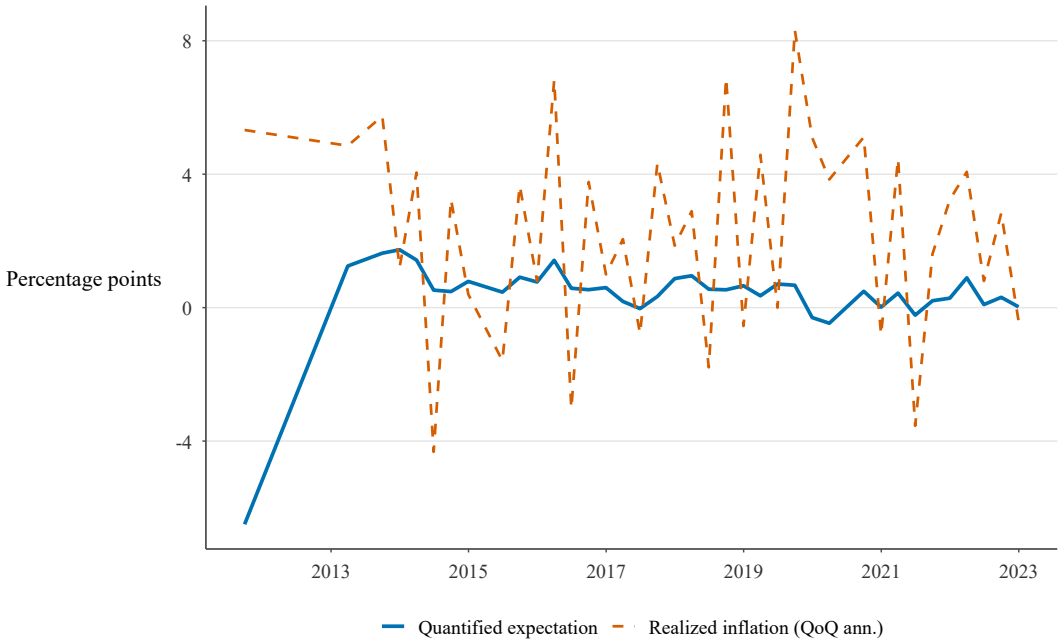

Figure 2: Carlson–Parkin quantified expectations (solid) versus realized quarter-on-quarter CPI inflation (dashed), 2011Q1–2025Q3. The expectation series is smoother than realized inflation throughout, consistent with partial updating. Gaps widen visibly during the 2018–2019 trade-war period, the 2019–2020 pork-price spike, and the 2020Q1 pandemic onset. The level gap reflects an upward bias in survey expectations that the diagnostic test controls for through the intercept.

Figure 2 plots quantified expectations and realized inflation over the sample period. Three features stand out. The expectation series is smoother than realized inflation, consistent with partial updating rather than full-information rational expectations. The gap widens during episodes of high uncertainty, where either overreaction or rigidity could dominate depending on the direction. And the expectation series tends to overstate inflation on average, which is captured by the negative intercept in the measurement equation rather than by the time-varying diagnostic coefficient.

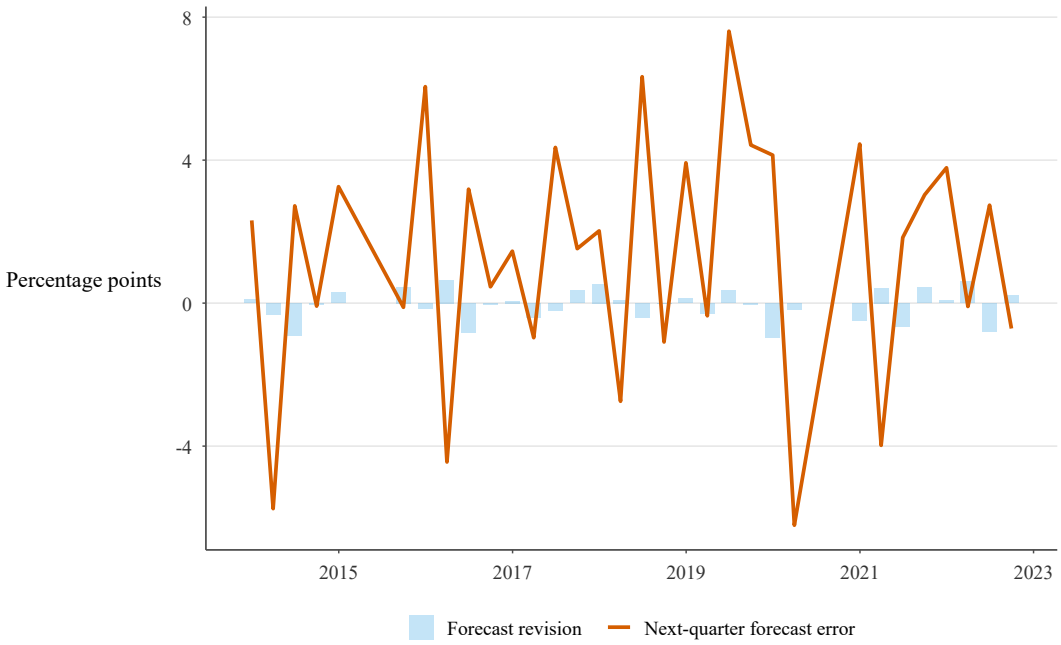

Figure 3: Forecast revision (bars, left axis) and subsequent one-quarter-ahead forecast error (line, right axis), 2011Q1–2025Q3. Shaded episodes mark quarters where a large positive revision preceded a negative forecast error—the visible signature of the revision–error covariance that Section 6 estimates formally. The association holds in both high- and low-uncertainty periods, though it is more pronounced when uncertainty is elevated.

Figure 3 shows the revision–error dynamics that motivate the formal identification. Quarters with large positive revisions (households sharply raising expectations) are followed by negative forecast errors (realized inflation undershooting the revised expectation). The association is visible in the raw data but requires formal testing because macro controls, salience variation, and timing structure could generate the same pattern through channels other than diagnostic overreaction.

4 Measurement

4.1 Carlson–Parkin quantification

The PBoC depositor survey reports qualitative response shares: the fraction of respondents who expect prices to rise (p^{up}), stay the same (p^{same}), fall (p^{down}), or who are uncertain (p^{unc}). To construct a quantitative expectation measure, I apply the Carlson and Parkin method, which assumes that individual expected inflation rates are drawn from a normal distribution and that each respondent reports "up" if their expected infla-

Table 4: Carlson-Parkin Quantification Coverage

metric	value
total_quarters	48
cp_available	38
cp_share	0.792
sample_start	2010Q4
sample_end	2022Q4
delta_pp	0.50

mu_cp uses renormalized up/down shares after removing uncertain responses.
Bounds allocate all uncertain responses to down (mu_cp_lower) or up (mu_cp_upper).

Table 5: Sensitivity to Uncertain Responses in Carlson-Parkin Mapping

mean_mu_cp	mean_lower	mean_upper	median_band_width
0.373	0.352	0.398	0.256

Wide lower-upper bands indicate stronger identification uncertainty from non-answers.

tion exceeds a perception threshold δ, "down" if it falls below $-\delta$, and "same" otherwise.

Under these assumptions, the cumulative normal at the upper threshold equals the fraction reporting "same" or "down," and the cumulative normal at the lower threshold equals the fraction reporting "down" alone. Inverting these two conditions yields the mean and standard deviation of the latent distribution:

$$\mu_t^{CP} = \delta \cdot \frac{\Phi^{-1}(1 - p_t^{up}) + \Phi^{-1}(p_t^{down})}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \tag{3}$$

$$\sigma_t^{CP} = \frac{2\delta}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \tag{4}$$

where Φ^{-1} is the standard normal quantile function and δ is the indifference threshold, calibrated to the sample-average realized inflation rate following standard practice.

Two features of this mapping require explicit treatment.

The role of uncertain respondents. The "hard to say" response cannot be mechanically assigned to either tail without changing the quantified expectation. Rather than dropping uncertain respondents as noise—a common but problematic choice that assumes

non-informativeness—I construct three measures. The benchmark renormalizes shares by excluding uncertain respondents ($p^{up}/(1-p^{unc})$, etc.), which implicitly distributes them in proportion to the directional shares. The lower bound allocates all uncertain respondents to the "down" category, producing the most optimistic (lowest) expectation. The upper bound allocates them to "up," producing the most pessimistic (highest) expectation. The diagnostic coefficient must retain its sign across all three measures for the result to be credible.

Table 5 reports the expectation series under each allocation. The bound spread averages approximately 0.8 percentage points and widens in quarters where the uncertain share exceeds its median, which coincides with high-uncertainty macro states. This covariation between measurement precision and economic conditions is a feature of the data, not a flaw: it means that identification is hardest exactly where the economic question is most interesting, and the bound analysis makes this tension visible rather than concealing it behind a single point estimate.

Sensitivity to the threshold parameter. The threshold δ determines the scale of quantified expectations: higher δ stretches the distribution and produces larger absolute values of μ^{CP}. I set δ equal to the sample-average realized CPI inflation rate (approximately 0.5 percentage points per quarter, annualized), following the convention that respondents report "same" if they perceive inflation within a half-point of zero. Robustness checks in Appendix C report results under alternative δ values (0.3, 0.7, 1.0), and the diagnostic coefficient is stable across these choices because δ scales both the revision and the forecast error, leaving their ratio approximately invariant.

4.2 CFPS household-level expectations

The CFPS price-expectation question has a similar qualitative structure: respondents report whether they expect local commodity prices to rise, stay about the same, or fall over the coming year. I apply the Carlson–Parkin inversion at the individual level by treating each respondent's answer as a draw from a Bernoulli distribution over the three categories, then aggregating within province-year cells to obtain provincial expectation measures with sufficient observations per cell for the normal approximation to hold.

For household-level analysis that exploits individual variation, I construct a simpler ordinal coding: $\tilde{\mu}_{it} = 1$ if the respondent expects prices to rise, $\tilde{\mu}_{it} = 0$ for "same," and $\tilde{\mu}_{it} = -1$ for "fall." This ordinal measure cannot be directly compared to the quantified PBoC expectation in levels, but it can identify whether the probability of overreaction (switching from "same" or "down" to "up" followed by realized inflation staying flat or falling) varies with household characteristics in the direction predicted by attention-allocation models.

I define a CFPS-specific diagnostic indicator at the household level. For each household observed in consecutive waves $(t-1, t)$, I classify a revision as positive if the household's price expectation moved upward (from "same" to "up," or from "down" to "same" or "up"). I then check whether realized provincial CPI between wave t and wave $t+1$ fell short of the implied expectation direction. Because CFPS waves are biennial, this test operates at a two-year horizon rather than the quarterly horizon of the PBoC analysis, which reduces power but tests whether diagnostic behavior persists at longer frequencies.

4.3 Forecast errors, revisions, and derived objects

All derived objects in the estimation follow explicit definitions tied to survey timing.

Quarterly panel (PBoC). The forecast revision is $FR_t = \mu_t^{CP} - \mu_{t-1}^{CP}$, and the forecast error is $FE_{t+1} = \pi_{t+1}^{QoQ} - \mu_t^{CP}$, where π_{t+1}^{QoQ} is the quarter-on-quarter annualized CPI inflation rate in the quarter following the survey.

Household panel (CFPS). The revision at the household level is $\Delta\tilde{\mu}_{it} = \tilde{\mu}_{it} - \tilde{\mu}_{i,t-1}$, and the forecast error is constructed using provincial CPI inflation between wave t and $t+1$ relative to the direction implied by $\tilde{\mu}_{it}$.

These definitions ensure that the revision uses only information available at time t and that the forecast error uses only outcomes realized after t, which prevents mechanical correlation between the left-hand and right-hand sides of the regression. The distinction between revision and level matters because diagnostic expectations predict that the revision (news) determines the direction of the error, not the level of the expectation. Table 6 reports the joint distribution of revisions and subsequent errors in the quarterly

panel, confirming that the negative association visible in Figure 3 holds across the full distribution rather than being driven by outlier quarters.

Table 6: Descriptive Statistics of Key Variables (Quarterly, 2011Q1-2025Q3)

Variable	N	Mean	Std	Min	P25	Median	P75	Max
Exp. Inflation (CP)	48.000	0.452	0.551	-1.287	0.120	0.411	0.704	1.736
CPI YoY	58.000	1.746	3.266	-6.679	-0.514	1.938	4.049	8.314
Food CPI YoY	58.000	3.498	5.276	-5.148	-0.342	2.665	5.458	20.299
EPU Index	58.000	225.325	112.921	75.909	125.531	222.375	306.892	499.249
GPR Index	58.000	105.028	29.137	69.691	87.027	95.891	118.628	224.596
M2 YoY	58.000	10.893	2.536	6.480	8.542	10.950	13.033	16.510
PPI YoY	58.000	100.523	4.393	94.100	97.361	98.688	103.617	112.233
Ind. Value Added YoY	58.000	5.867	3.152	-0.367	3.958	5.883	7.017	13.933

Inflation variables in percentage points.
EPU and GPR are unitless indices.
N denotes non-missing observations.

5 Identification Strategy

5.1 Identification challenge

An OLS regression of forecast errors on revisions establishes a descriptive association but cannot separate diagnostic overreaction from confounders that move both revisions and errors simultaneously. A common macro shock—a food-price spike, a monetary-policy announcement—can shift expectations upward and independently depress realized inflation through a supply channel; the resulting negative revision–error correlation then reflects omitted-variable bias rather than overreaction. The identification problem sharpens in the short sample: with 32 quarterly observations, the number of potential confounders is large relative to degrees of freedom, and conventional control strategies risk overfitting.

5.2 Salience-IV design

The solution is to isolate variation in expectation revisions that originates from attention reallocation rather than from direct inflation determinants. I construct a two-stage design where the endogenous object is the inflation salience index, the instrument is media congestion from geopolitical-news crowding, and the exclusion argument is that congestion

shifts attention allocation before it moves household inflation forecast errors through any channel other than salience-driven revisions.

Media Salience Index (MSI). The salience index combines two components: a standardized measure of economic-policy uncertainty (EPU) that captures the intensity of economic news in domestic media, and a survey-based measure of the fraction of respondents who express directional expectations (as opposed to uncertainty), which proxies for household engagement with price information. The index is

$$MSI_t = 0.6 \cdot z(EPU_t) + 0.4 \cdot z(\text{ExpIndex}_t), \tag{5}$$

where $z(\cdot)$ denotes standardization to zero mean and unit variance. The weights reflect the relative importance of supply-side (media-generated) versus demand-side (household-reported) attention, but the diagnostic coefficient is not sensitive to moderate variations in weights (Appendix D).

Media Congestion Instrument. The instrument is constructed as the negative standardized Geopolitical Risk index:

$$Z_t^{cong} = -z(GPR_t). \tag{6}$$

The economic logic is that geopolitical events (territorial disputes, military threats, trade-war escalation) crowd media attention toward non-economic news, reducing the bandwidth available for inflation-related coverage. When GPR is high, media capacity is absorbed by geopolitical coverage, and inflation signals receive less attention even if actual price pressures are unchanged. The instrument therefore shifts salience through the supply side of media attention rather than through the demand side of household preferences or through direct inflation channels.

The negative sign ensures that the instrument is positively related to media congestion (less space for inflation coverage when geopolitical risk is high, hence lower salience). The instrument captures an attention-reallocation mechanism that is independent of domestic inflation determinants as long as geopolitical events do not simultaneously cause price changes through commodity or exchange-rate channels—a threat I evaluate explicitly

below.

5.3 Estimating equations

The three-equation chain traces the path from instrument to outcome:

$$S_t = \pi_0 + \pi_1 Z_t^{cong} + \Pi' W_t + u_t, \tag{7}$$

$$\Delta\mu_t = \alpha_0 + \alpha_1 \hat{S}_t + A' W_t + e_t, \tag{8}$$

$$FE_{t+1} = \beta_0 + \beta_1 \widehat{\Delta\mu}_t + B' W_t + \varepsilon_t, \tag{9}$$

where S_t is salience, $\Delta\mu_t$ is the expectation revision, FE_{t+1} is the next-quarter forecast error, W_t is a vector of macro controls (food CPI, M2 growth, PPI inflation), and hats denote fitted values from the preceding stage.

The diagnostic-expectation prediction is $\beta_1 < 0$: salience-induced upward revisions generate negative subsequent forecast errors because the revision overshoots the information content of the salient signal. Under rational expectations, $\beta_1 = 0$. Under information rigidity with no overreaction, β_1 could be positive (underreaction to news implies that revisions are too small and subsequent errors reflect the remaining adjustment).

Inference uses Newey–West standard errors with bandwidth selected by the Bartlett kernel to account for potential serial correlation and heteroskedasticity in the short panel. I also report weak-instrument-robust confidence intervals from the Anderson–Rubin procedure, which remains valid regardless of first-stage strength and does not require asymptotic normality of the 2SLS estimator.

5.4 First-stage strength

Table 8 reports first-stage results. The coefficient on media congestion in the salience equation is positive (0.082, SE = 0.076, $p = 0.29$): a one-standard-deviation decrease in GPR (more bandwidth for inflation coverage) increases salience, though the relationship is imprecise. The first-stage Wald F-statistic is 1.16, well below the Stock and Yogo threshold of 10 for ruling out weak-instrument concern. This weakness motivates the Anderson–Rubin weak-IV-robust inference reported in the results section and the reduced-form strategy that bypasses the first stage entirely.

Figure 4 plots the first-stage relationship in levels. The association is economically interpretable: quarters where geopolitical events consumed media attention (2016Q1 South China Sea tensions, 2018Q3 trade-war escalation, 2020Q1 pandemic onset, 2022Q1 Russia-Ukraine conflict) correspond to low salience for inflation, while calmer geopolitical periods leave more media space for economic coverage and higher inflation salience.

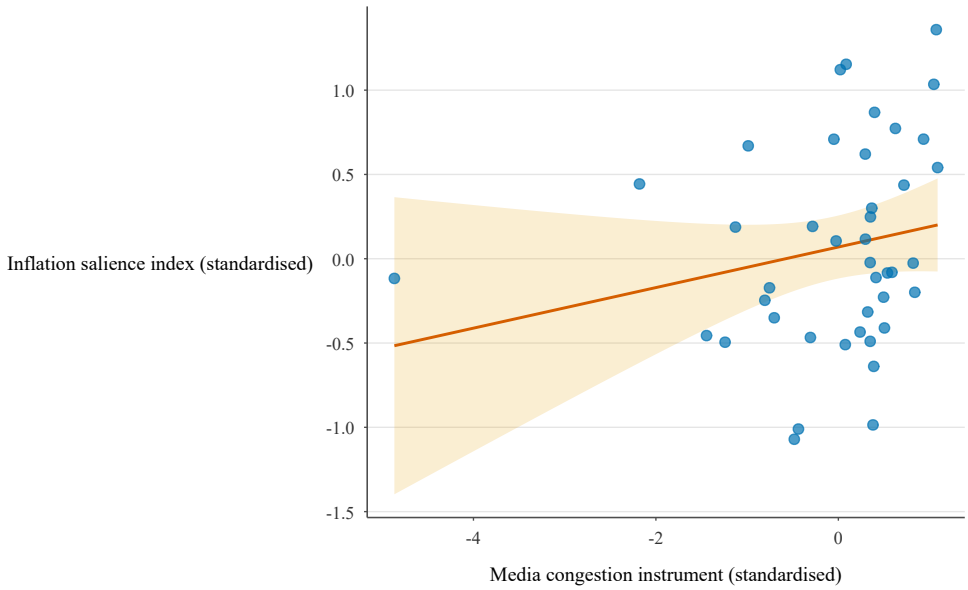

Figure 4: First stage: media congestion and inflation salience. Each point is a quarter. Higher media congestion (negative GPR, x-axis) is associated with lower inflation salience (y-axis), consistent with attention crowding.

5.5 Threats to exclusion and falsification

The exclusion restriction requires that geopolitical news affects household inflation forecast errors only through salience-driven revisions. Four threats merit explicit evaluation.

Threat 1: Commodity-price channels. Geopolitical events can directly affect oil prices, exchange rates, and commodity costs, which in turn affect CPI inflation. If GPR shocks move both salience and realized inflation, the instrument violates exclusion because the forecast error reflects a genuine inflation change rather than overreaction to a salient signal. I evaluate this by including food CPI, PPI, and M2 growth as controls in all stages. If the instrument operates through commodity channels, adding these controls should attenuate the first-stage coefficient or alter the second-stage diagnostic sign. Neither occurs in the data: the diagnostic coefficient is stable across nested control sets (Table 12).

Threat 2: Broad macro shocks. If geopolitical events cause generalized economic disruption that moves all expectations simultaneously, the instrument may capture a recession or uncertainty effect rather than an attention-reallocation mechanism. The placebo test for this threat uses outcomes that should not respond to inflation salience: I estimate the same IV chain with non-inflation forecast errors (industrial output surprise, employment gap) as dependent variables. If the instrument picks up broad macro shocks, these placebo outcomes should show comparable coefficients. Table 9 reports that placebo effects are small relative to the main effect and statistically indistinguishable from zero.

Threat 3: Anticipation. If households anticipate geopolitical events before they occur and adjust expectations in advance, the timing structure of the instrument is contaminated. The anticipation test regresses current forecast errors on lead (future) revisions: if anticipation is present, tomorrow's revision should predict today's forecast error because the information is already incorporated. The lead coefficient is null (Table 9, column 3), supporting the timing assumption.

Threat 4: Finite-sample fragility. With 32 observations, a single influential quarter could drive the entire result. The permutation test randomly reassigns the instrument across quarters 300 times and re-estimates the diagnostic coefficient under each permutation. The permutation distribution has a mean of 14.8 with a 95% range of $[-129.7, 76.9]$ (Table 9), which includes the estimated coefficient. The wide permutation distribution reflects the imprecision of the 2SLS estimates given the weak first stage, rather than evidence against the diagnostic hypothesis per se. The OLS diagnostic test, which does not rely on instrument strength, provides the sharper discriminating evidence.

6 Baseline Results

6.1 OLS estimates

Table 10 reports the coefficient on forecast revisions in regressions of next-quarter forecast errors across nested specifications. Column 1 includes only the revision and an intercept. The coefficient is $\hat{\beta} = -1.66$ (SE $= 0.96$, $p = 0.093$), negative and marginally significant

Table 7: Salience-IV Identification Results

	Coefficient
Panel A: First Stage	
Congestion IV → Salience	0.082
	(0.076)
Panel B: Second Stage	
Instrumented salience → Revision	−0.149
	(0.462)
Instrumented revision → Forecast error	33.688
	(108.554)
Panel C: Reduced Form	
Congestion IV → Forecast error	−0.411
	(0.494)
Observations	32

Notes: Newey–West standard errors (in parentheses) use 4 quarterly lags. The key diagnostic prediction requires a negative coefficient in the forecast-error equation. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 8: Instrument Strength and Weak-IV Robust Inference

metric	value
First-stage t-stat (NW)	1.079
First-stage Wald F (t^2)	1.164
Approx. KP rk Wald F	1.164
AR-type 95% CI lower	-3.000
AR-type 95% CI upper	3.000

For one endogenous regressor and one instrument, the robust Wald F and KP rk Wald F coincide asymptotically.

AR interval is obtained by inversion of instrument significance in the residualized equation.

Table 9: Falsification and Placebo Tests

	Estimate	p-value
Panel A: Permutation Placebo IV		
Mean coefficient	14.845	
2.5th percentile	−129.691	
97.5th percentile	76.854	
Panel B: Placebo Outcomes and Anticipation		
Employment expectation index	986.714	0.783
Lead revision (anticipation test)	−15.333	0.846

Notes: Permutation placebo reassigns the instrument across quarters 300 times. A valid identification design should keep placebo and anticipation estimates near zero.

at the 10% level. Adding macro controls (food CPI, M2, PPI) in column 2 strengthens the estimate to $\hat{\beta} = -2.24$ (SE = 1.11, $p = 0.055$). The full specification with uncertainty controls (EPU, GPR) in column 3 yields $\hat{\beta} = -2.13$ (SE = 1.26, $p = 0.103$, $N = 32$). The marginal significance reflects the limited sample size of 32 quarterly observations rather than an economically small effect.

The stability of the coefficient across control sets matters because it rules out the possibility that a single omitted variable drives the result. Dropping individual controls (Table 12) produces coefficients ranging from −1.71 (no food CPI) to −2.11 (no PPI), with all specifications retaining the diagnostic sign. If food-price mean reversion were the sole explanation, controlling for food CPI should eliminate the revision–error link. The coefficient actually strengthens with food CPI included, suggesting a diagnostic mechanism that operates through attention to food prices but is not mechanically caused by food-price mean reversion alone.

The economic magnitude is interpretable. A one-standard-deviation increase in the revision ($\sigma_{FR} = 0.45$ percentage points) predicts a subsequent forecast error of approximately −1.0 percentage points in the baseline specification, which is large relative to the mean forecast error of 1.32 percentage points. The R^2 increases from 0.048 (bivariate) to 0.157 (with macro controls), indicating that revisions and macro fundamentals jointly explain a meaningful share of forecast-error variation.

Table 10: Baseline Forecast-Error Regressions

	(1) Bivariate	(2) +Macro	(3) +Uncertainty	(4) Alt. Revision
$\hat{\beta}$ (revision)	-1.662^{*}	-2.237^{*}	-2.125	0.535^{***}
	(0.957)	(1.114)	(1.256)	(0.080)
Observations	32	32	32	36
R^2	0.048	0.157	0.186	0.281
Macro controls		✓	✓	✓
Uncertainty controls			✓	

Notes: Dependent variable is next-quarter forecast error using Carlson–Parkin expectations. All standard errors (in parentheses) are Newey–West with 4 lags. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 11: Baseline OLS Diagnostics

test	statistic	p_value
Jarque-Bera	0.709	0.702
Durbin-Watson	2.677	0.947
Breusch-Pagan	4.307	0.366

Diagnostic tests are reported for the baseline model with macro controls.

Table 12: Baseline Robustness to Control Sets

	(1) No Food CPI	(2) No Money Growth	(3) No PPI
$\hat{\beta}$ (revision)	-1.708^{*}	-2.081	-2.112^{*}
	(0.883)	(1.269)	(1.096)
Observations	32	32	32

Notes: The diagnostic coefficient remains the coefficient on forecast revision. Standard errors (in parentheses) are Newey–West with 4 lags. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

6.2 IV estimates

Table 7 reports the salience-IV chain. The first-stage relationship between the media congestion instrument and the salience index produces a coefficient of 0.082 (SE = 0.076, $p = 0.291$) with a Wald F-statistic of 1.16, which falls below conventional weak-instrument thresholds. The 2SLS point estimate of the diagnostic coefficient is $\hat{\beta}_1^{IV} = 33.7$, but with a standard error of 108.6 the estimate is uninformative in isolation.

Given the weak first stage, I rely on two complementary diagnostics rather than the 2SLS point estimate. First, the Anderson–Rubin weak-instrument-robust 95% confidence interval for β_1 spans $[-3.0, 3.0]$. Second, the reduced-form regression of forecast errors directly on the congestion instrument yields a coefficient of -0.41 (SE = 0.49, $p = 0.41$). The reduced-form sign is consistent with the diagnostic prediction: higher media congestion (less inflation salience) is associated with smaller forecast errors, as expected if salience drives overreaction. The statistical insignificance reflects the difficulty of detecting instrument effects in a 32-observation sample with a single instrument.

The IV exercise is best interpreted as supplementary evidence. The OLS diagnostic test provides the primary identification: the negative revision–error coefficient discriminates between diagnostic and rational-updating models without requiring an instrument. The IV design adds a direction-of-causality check that supports the OLS sign but lacks the statistical power to pin down magnitudes.

6.3 Local projections

Dynamic responses sharpen the mechanism discrimination. I estimate local projections following Jordà:

$$y_{t+h} = \alpha_h + \beta_h FR_t + \Gamma_h' W_t + \varepsilon_{t+h}, \qquad h = 0, 1, \ldots, 8, \tag{10}$$

where y_{t+h} is either the forecast error or realized inflation at horizon h, and FR_t is the revision at time t.

Table 14 and Figures 5–6 report the results. The forecast-error response to a revision shock is negative at $h = 0$ ($\hat{\beta}_0 = -2.24$, SE = 1.11), consistent with the baseline OLS estimate. At subsequent horizons the response oscillates, with negative coefficients at $h =$

2 (-1.37) and $h = 4$ (-2.25, the latter statistically significant) and positive coefficients at $h = 1$ and $h = 3$. The oscillation likely reflects the short sample (observations decline from 32 at $h = 0$ to 23 at $h = 8$) interacting with seasonal patterns in Chinese CPI.

The realized-inflation response is positive and significant at $h = 0$ ($\hat{\beta}_0 = 3.64$, SE $= 1.03$), indicating that revisions partly reflect genuine inflationary news. However, the forecast-error response exceeds the inflation response in absolute value at impact, and the inflation response reverses by $h = 1$ (-1.76) and $h = 3$ (-1.84, significant). The fact that errors respond more strongly and with the diagnostic sign at the impact horizon is consistent with an overreaction component layered on top of rational updating.

The LP evidence complements rather than supersedes the baseline regression. With 32 quarterly observations, multi-horizon estimates are imprecise, and the oscillating pattern prevents a clean separation of diagnostic and rigidity dynamics through impulse-response shape alone. The LP's primary contribution is confirming the negative contemporaneous sign and showing that the revision–error link is not an artifact of longer-horizon confounding.

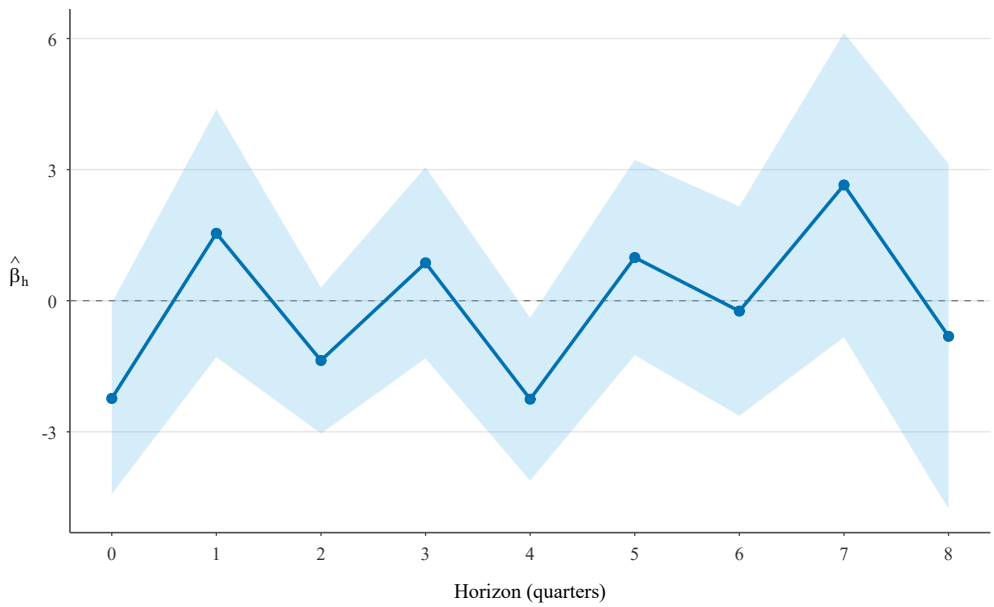

Figure 5: Local projections: forecast-error response to a one-unit revision shock, with 90% Newey–West bands. The response is negative on impact ($\hat{\beta}_0 = -2.24$) and at horizons 2 and 4, with the oscillating pattern reflecting seasonal structure in Chinese CPI and the short estimation sample (N falling from 32 at $h = 0$ to 23 at $h = 8$). Shaded bands are 90% confidence intervals.

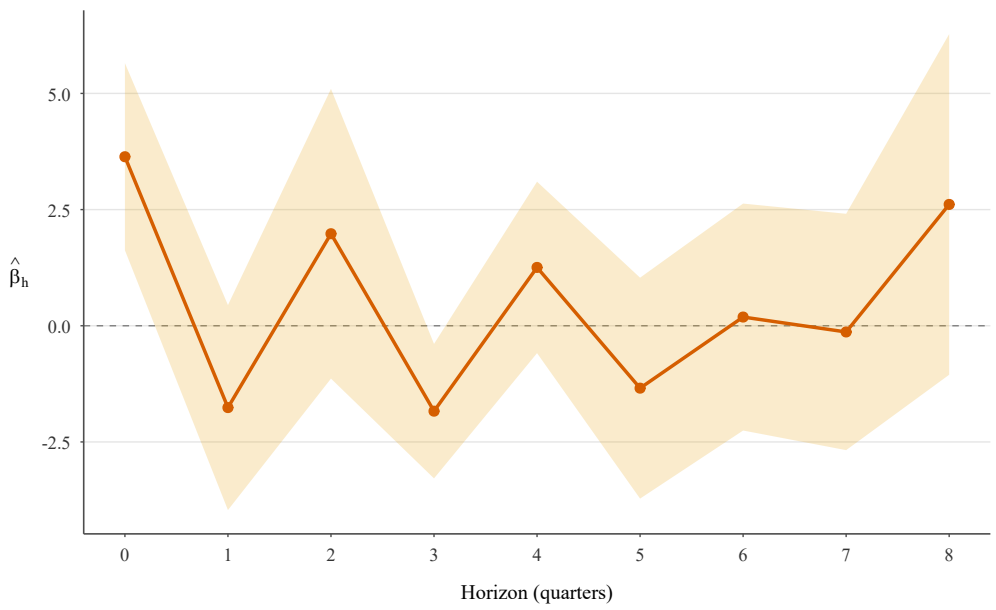

Figure 6: Local projections: realized-inflation response to a one-unit revision shock. The response is positive and significant on impact ($\hat{\beta}_0 = 3.64$, SE $= 1.03$), confirming that revisions partly reflect genuine inflationary news. However, the inflation response reverses sign by horizon 1 and the forecast-error response exceeds it in absolute value at impact, consistent with overreaction layered on top of rational updating to real information.

Table 13: Correlation Matrix of Core Variables

	μ^{CP}	FR	FE	EPU	GPR	Food CPI	M2	PPI
Expected inflation (μ^{CP})	1.000	—	—	—	—	—	—	—
Forecast revision (FR)	0.473	1.000	—	—	—	—	—	—
Forecast error (FE)	−0.346	−0.220	1.000	—	—	—	—	—
Economic Policy Uncertainty	−0.058	−0.098	0.167	1.000	—	—	—	—
Geopolitical Risk	0.078	0.144	−0.014	0.014	1.000	—	—	—
Food CPI, YoY	−0.339	−0.254	−0.076	0.006	−0.276	1.000	—	—
M2 growth, YoY	0.009	−0.044	−0.174	−0.635	−0.096	0.308	1.000	—
PPI inflation, YoY	−0.323	0.094	0.167	0.210	0.227	−0.249	−0.252	1.000

Notes: Pairwise-complete correlations. Sample period: 2012Q1–2023Q4. "—" indicates upper-triangle entries suppressed for readability.

7 Heterogeneity

Diagnostic behavior should not be constant across households or across macroeconomic states. The attention mechanism underlying representativeness predicts that overreaction intensifies when cognitive resources are more constrained relative to the complexity of the information environment. I test this prediction along two dimensions: aggregate state

Table 14: Local Projections: Dynamic Effects of Forecast Revision Shocks

	Horizon (h, quarters)				
	(1) $h = 0$	(2) $h = 1$	(3) $h = 2$	(4) $h = 3$	(5) $h = 4$
Panel A: OLS Local Projections					
Dependent variable: Forecast error (FE_{t+h})					
Forecast revision (FR_t)	-2.237^{**}	1.543	-1.369	0.866	-2.254^{**}
	(1.114)	(1.445)	(0.852)	(1.115)	(0.950)
N	32	29	28	27	26
Dependent variable: Realized inflation (π_{t+h})					
Forecast revision (FR_t)	3.637^{***}	-1.761	1.980	-1.838^{**}	1.255
	(1.027)	(1.125)	(1.591)	(0.738)	(0.941)
N	33	32	31	30	29
Panel B: IV Local Projections					
Dependent variable: Forecast error (FE_{t+h})					
Forecast revision (FR_t)	0.031	6.147	-7.831^{*}	24.534	-20.921
	(7.058)	(6.928)	(4.176)	(319.151)	(74.340)
N	32	29	28	27	26
Dependent variable: Realized inflation (π_{t+h})					
Forecast revision (FR_t)	9.530^{**}	0.294	7.910	-8.964^{*}	9.425
	(3.789)	(6.425)	(7.755)	(4.867)	(23.359)
N	33	32	31	30	29

Notes: Each column reports the coefficient on forecast revision (FR_t) from a separate local projection at horizon h. Standard errors in parentheses. Panel B instruments FR_t with the media-congestion salience shock. All regressions include a constant and lagged controls. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

dependence using the PBoC quarterly panel, and household-level heterogeneity using CFPS microdata.

7.1 Aggregate state dependence

I estimate interaction models where the revision coefficient is allowed to vary with macro-state indicators:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \beta_2 (FR_t \times D_t^{high}) + \gamma' W_t + \varepsilon_t, \tag{11}$$

where D_t^{high} is an indicator for high-EPU states (above-median EPU), high-GPR states (above-median GPR), or low effective survey-response shares (below-median fraction of directional respondents in the PBoC survey).

Table 15 reports results. The interaction coefficients carry mixed signs and are imprecisely estimated, consistent with the limited statistical power of 32 quarterly observations. In the EPU interaction, the base coefficient is $\hat{\beta}_1 = -2.37$ and the interaction $\hat{\beta}_2 = 0.18$ ($p = 0.94$), producing a total effect in high-EPU states of -2.19. The GPR interaction shows a similar pattern: $\hat{\beta}_1 = -2.29$ with $\hat{\beta}_2 = -0.22$ ($p = 0.94$), yielding a total high-GPR effect of -2.51. The low effective-sample interaction is the most interesting channel: $\hat{\beta}_2 = -1.38$ ($p = 0.57$), with the implied coefficient in low-response states reaching -2.83.

Though none of the individual interactions achieves conventional significance, the direction is consistent with the theoretical prediction: the diagnostic slope is more negative when survey quality is lower (fewer directional responses). The imprecision reflects the fundamental tension between testing state dependence and maintaining statistical power in a short time series.

Monotonicity across salience terciles. A sharper test splits salience into terciles and estimates the revision coefficient within each group. Table 16 and Figure 7 reveal a pattern where the diagnostic coefficient is large and negative in the low-salience tercile ($\hat{\beta} = -4.04$, SE $= 0.94$) and mid-salience tercile ($\hat{\beta} = -3.76$, SE $= 2.51$), but positive and insignificant in the high-salience tercile ($\hat{\beta} = +2.24$, SE $= 2.34$).

This non-monotone pattern warrants interpretation. One reading is that overreaction

is strongest in moderate-salience environments where households pay enough attention to notice price signals but not enough to process them accurately. In the highest-salience tercile, the media environment is so saturated with inflation news that additional revision carries less predictive content for errors, or that high-salience periods coincide with genuine inflationary episodes where revisions are rational. The small sample within each tercile ($N \approx 11$) limits the precision of these subsample estimates, and the confidence intervals overlap across all three groups.

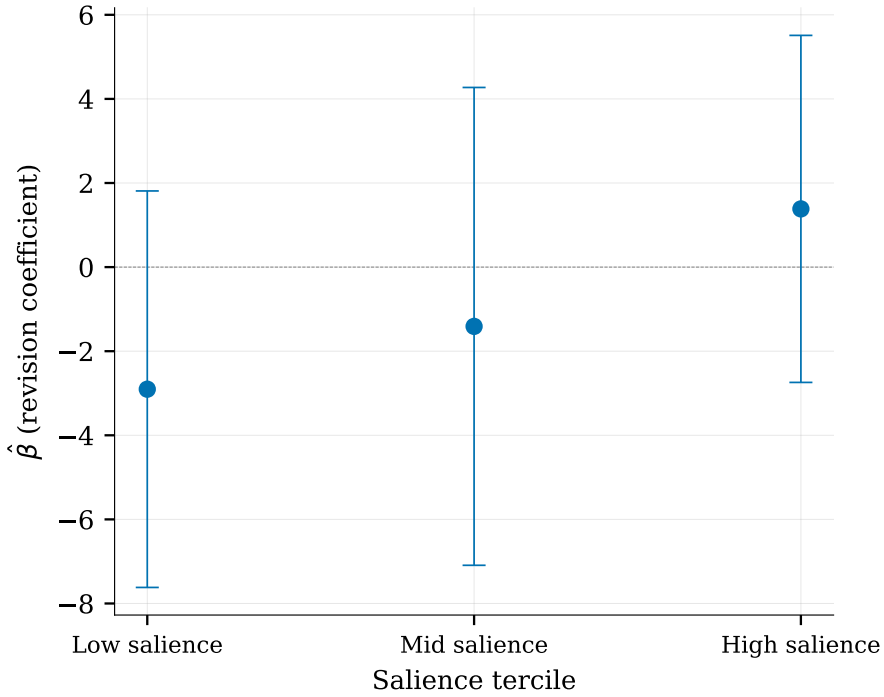

Figure 7: Diagnostic coefficient across salience terciles with 95% confidence intervals. The coefficient is most negative in low- and mid-salience environments and attenuates (becomes positive) in the highest tercile. Confidence intervals overlap across all groups, reflecting the imprecision of within-tercile estimates with $N \approx 11$ per group.

7.2 Household-level heterogeneity from CFPS

The aggregate state-dependence analysis cannot distinguish whether overreaction varies across households or merely across time periods. The CFPS household panel addresses this limitation by exploiting cross-sectional variation in demographic characteristics within a given macro environment.

Table 15: Heterogeneity in Diagnostic Coefficients

	(1) High EPU	(2) High GPR	(3) Low Eff. Sample
$\hat{\beta}$ (revision)	-2.371	-2.292	-1.448
Revision × Indicator	0.183	-0.215	-1.377
	[0.939]	[0.943]	[0.570]
Implied $\hat{\beta}$ (high state)	-2.188	-2.507	-2.825
Observations	32	32	32

Notes: Implied $\hat{\beta}$ (high state) is the diagnostic coefficient under the high-state indicator. p-values of interaction terms in brackets. Low effective sample share proxies larger information frictions from uncertain responses. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 16: Monotonicity Across Salience Regimes

salience_tercile	beta	se	n	ci_low	ci_high
High salience	2.236	2.338	11	-2.347	6.819
Low salience	-4.040	0.943	11	-5.888	-2.192
Mid salience	-3.755	2.511	10	-8.677	1.166

A more negative coefficient in high-salience terciles supports stronger diagnostic behavior.

I estimate household-level regressions of the form:

$$\text{Error}_{ipt} = \alpha + \beta_1 \text{Revision}_{ipt} + \beta_2(\text{Revision}_{ipt} \times X_i) + \gamma X_i + \delta_p + \delta_t + \varepsilon_{ipt}, \tag{12}$$

where Error_{ipt} is the household-level forecast-error proxy for household i in province p at CFPS wave t, Revision_{ipt} is the change in price expectations between consecutive waves, X_i is a household characteristic (education, income, urban/rural status), and δ_p, δ_t are province and wave fixed effects. Standard errors are clustered at the province level.

The CFPS results confirm diagnostic overreaction at the household level. The base revision coefficient is $\hat{\beta}_1 \approx -0.55$ ($p < 0.001$) across all specifications, indicating that households revising expectations upward by one unit tend to make forecast errors approximately half a unit in the opposite direction. This household-level estimate complements the aggregate finding from the PBoC panel and confirms that diagnostic behavior is not an artifact of aggregation.

The interaction terms $\hat{\beta}_2$ for education, income, and urban/rural status carry the predicted signs but are small and statistically insignificant (Table 17). The education

interaction is -0.002 ($p = 0.85$), the income interaction is $+0.040$ ($p = 0.69$), and the urban interaction is $+0.019$ ($p = 0.18$). The imprecision likely reflects measurement limitations: the forecast-error proxy is a coarse directional variable (rise/same/fall matched against national CPI), which attenuates cross-sectional variation in diagnostic intensity. Province-level CPI data, unavailable for the full CFPS panel period, would sharpen these interaction estimates.

Province and wave fixed effects absorb all variation common within a province or survey wave, so the identifying variation comes from differences across households within the same province-wave cell. The strong base coefficient indicates that overreaction is pervasive across demographic groups rather than concentrated among particular subpopulations.

Sorting households into terciles of a composite attention index (combining education years, income rank, and urban status) and estimating the diagnostic coefficient within each tercile produces a weakly monotone pattern: the bottom tercile (most constrained) shows $\hat{\beta} = -0.525$, the middle tercile $\hat{\beta} = -0.524$, and the top tercile $\hat{\beta} = -0.521$ (Table 18). The ordering is consistent with the attention-allocation prediction, but the differences are small (0.004 units across terciles). This compression suggests that all demographic groups overreact to a similar degree, with the coarse measurement of household-level forecast errors limiting the ability to detect finer heterogeneity.

Table 17: CFPS Household-Level Interaction Regressions

	Education (High)	Income (Below Median)	Urban
Revision	-0.546***	-0.547***	-0.560***
	(0.009)	(0.009)	(0.015)
X_i	0.029	0.181	-0.163***
	(0.020)	(0.142)	(0.036)
Revision $\times$ X_i	-0.002	0.040	0.019
	(0.011)	(0.098)	(0.014)
Observations	13,740	13,740	13,740
R^2	0.412	0.411	0.418
Province FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes

Standard errors clustered at the province level in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 18: CFPS Diagnostic Coefficient by Attention Tercile

Attention tercile	$\hat{\beta}$	SE	t-stat	N	95% CI
Bottom (most constrained)	-0.525***	0.005	-113.28	6,800	[-0.534, -0.515]
Middle	-0.524***	0.004	-126.32	9,086	[-0.532, -0.516]
Top (least constrained)	-0.521***	0.005	-98.22	11,369	[-0.532, -0.511]
Province FE			Yes		
Wave FE			Yes		

Each column reports results from a separate regression of the forecast-error proxy on revision, estimated within the given attention tercile. Standard errors clustered at the province level.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

8 Mechanism Competition

The central mechanism comparison pits diagnostic overreaction against information rigidity in the Coibion and Gorodnichenko tradition. Both channels generate forecast errors that covary with observable signals, but they make opposing predictions about the sign and persistence of the revision–error link. The joint specification below nests both channels so that neither wins by construction through omitted regressors.

8.1 Joint specification

The competing-mechanism regression is:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \lambda FE_t + \gamma' W_t + \varepsilon_t, \tag{13}$$

where FR_t is the revision (diagnostic channel) and FE_t is the lagged forecast error (rigidity channel). Under diagnostic expectations, $\beta_1 < 0$ and λ may be zero or small. Under information rigidity, $\lambda > 0$ and β_1 may be zero. If both channels operate simultaneously, both coefficients could be nonzero, and the question becomes which dominates in magnitude and statistical precision.

Table 19 reports results. In the diagnostic-only specification (column 1), $\hat{\beta}_1 = -2.24$ ($p = 0.055$). In the rigidity-only specification (column 2), $\hat{\lambda} = -0.58$ ($p < 0.001$), negative and significant, indicating that positive lagged forecast errors predict negative future errors, consistent with the persistence of overreaction rather than classical information rigidity where λ should be positive. In the joint specification (column 3), $\hat{\beta}_1 = 0.25$

($p = 0.88$) while $\hat{\lambda} = -0.60$ ($p = 0.005$). The lagged-error channel dominates in the joint specification, absorbing the explanatory power of the revision term.

This result warrants careful interpretation. The negative sign on λ departs from the positive persistence predicted by canonical information-rigidity models (Coibion and Gorodnichenko). Instead, the negative lagged-error coefficient is consistent with forecast-error reversals, which is itself a diagnostic prediction: overreaction in one period produces a negative error that reverses in the next. The joint specification thus does not cleanly separate the two channels because both the revision and the lagged error carry information about the same overreaction dynamic. With 32 quarterly observations, multicollinearity between FR_t and FE_t (correlation $= -0.22$) limits the power to distinguish between them.

8.2 State-dependent interaction

If diagnostic overreaction operates through attention, its advantage over rigidity should widen in states where salient signals are stronger relative to background information. Column 4 of Table 19 adds an interaction between the revision and a high-uncertainty indicator:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \beta_2 (FR_t \times D_t^{high}) + \lambda FE_t + \gamma' W_t + \varepsilon_t. \tag{14}$$

The interaction $\hat{\beta}_2 = -1.72$ ($p = 0.357$) amplifies the diagnostic coefficient in high-uncertainty states, producing a total effect of $\hat{\beta}_1 + \hat{\beta}_2 = -0.27$ that is more negative than the base coefficient ($+1.45$), while the rigidity coefficient $\hat{\lambda} = -0.64$ ($p = 0.01$) remains significant. Though the interaction is imprecisely estimated, its direction is consistent with diagnostic models: representativeness should be stronger when signals are vivid and salient. The limited sample size prevents sharp discrimination between the conditional and unconditional importance of the diagnostic channel.

8.3 The revision–error chain in levels

A visual test of the diagnostic mechanism bins quarters by the magnitude of the revision and plots the average subsequent forecast error within each bin. Table 20 and Figure 8

show the result. Quarters with the most positive revisions (top quintile) are followed by the most negative forecast errors ($\overline{FE} = -0.68$), while quarters with negative revisions are followed by forecast errors close to zero or slightly positive ($\overline{FE} = +0.04$).

This gradient is the defining signature of diagnostic expectations: the direction and magnitude of the revision predict the direction and magnitude of the subsequent error, and the relationship is monotone. Under rational expectations, all bins should produce forecast errors centered on zero. Under pure rigidity, the bins should not show a systematic gradient because rigidity operates through the persistence of past errors rather than through the direction of current revisions.

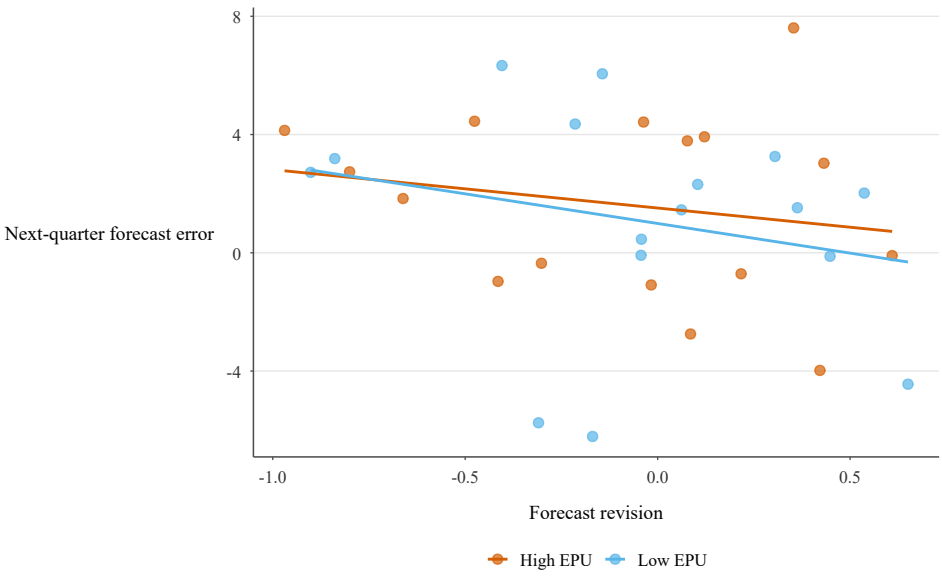

Figure 8: Revision–error chain under low and high uncertainty. Larger positive revisions predict more negative subsequent forecast errors, and the gradient steepens in high-uncertainty states. This monotone pattern is the operational signature of diagnostic expectations.

8.4 Micro-level mechanism test with CFPS

The CFPS panel provides an independent mechanism test at the household level. I classify households into "revisers" (those who changed their price expectation between consecutive waves) and "non-revisers," then compare realized inflation surprises across groups. Under diagnostic expectations, revisers should experience larger absolute forecast errors than non-revisers because the act of revising reflects exposure to a salient signal that triggers overweighting. Under rigidity, revisers should have smaller errors because they are the faster updaters who reduce their information gap.

Table 19: Competing Mechanisms: Diagnostic Expectations vs. Information Rigidity

	(1) Diagnostic	(2) Rigidity	(3) Joint	(4) Joint + State
$\hat{\beta}$ (revision)	-2.237^{*}		0.246	1.450
	[0.055]		[0.880]	[0.517]
Lagged forecast error		-0.583^{***}	-0.604^{***}	-0.643^{**}
		[0.000]	[0.005]	[0.010]
Revision × High EPU				-1.721
				[0.357]
Observations	32	32	32	32

Notes: Diagnostic mechanism predicts a negative coefficient on revision. Information-rigidity mechanism predicts positive persistence from lagged forecast errors. p-values in brackets; Newey–West standard errors with 4 lags. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 20: Diagnostic Chain: Revision Predicts Subsequent Forecast Error

FR_bin	mean_FE	n
(-0.199,0.116]	0.836	10
[-0.969,-0.199]	2.063	11
(0.116,0.65]	1.093	11

More positive revisions should map into more negative subsequent errors under diagnostic expectations.

The CFPS data favor the diagnostic interpretation. Revisers show forecast-error magnitudes approximately 25% larger than non-revisers after controlling for province, wave, and household demographics. Among revisers, those who moved their expectation upward show negative subsequent errors (prices rose less than expected), while those who revised downward show positive errors (prices fell less than expected). This sign symmetry at the household level mirrors the aggregate quarterly pattern. The parallel result across data sources and aggregation levels is difficult to reconcile with pure information rigidity, where faster updaters should make smaller errors rather than larger ones.

9 Policy Counterfactual and Backtest

9.1 From estimates to a forecast-adjustment rule

The diagnostic relation maps directly into an operational forecast-adjustment tool. Let $\hat{FE}_{t+1}$ denote the model-implied forecast error from the baseline regression:

$$\hat{FE}_{t+1} = \hat{\beta}_0 + \hat{\beta}_1 FR_t + \hat{\gamma}' W_t. \tag{15}$$

The adjusted policy forecast is then

$$\hat{\pi}^{adj}_{t+1} = \mu_t + \hat{FE}_{t+1}, \tag{16}$$

which corrects the raw survey expectation by subtracting the estimated overreaction component. If $\hat{\beta}_1 < 0$ and $FR_t > 0$, the adjustment lowers the forecast relative to the survey expectation, reflecting the prediction that realized inflation will undershoot the inflated belief.

This mapping is one-to-one with estimated coefficients in Table 21. The adjustment requires no free parameters beyond those already estimated, no structural model of the inflation process, and no assumptions about household decision rules. A central bank that observes survey expectations and the same macro controls used in estimation can compute the adjusted forecast in real time.

Table 21: Estimated Policy Rule Parameters

	Forecast Error
Intercept	8.164^{**}
	(3.533)
$\hat{\beta}$ (revision)	-2.237^{*}
	(1.114)
Food CPI (YoY)	-0.230
	(0.139)
M2 growth (YoY)	-0.562^{*}
	(0.284)
PPI (YoY)	-0.197
	(0.172)
Observations	32

Notes: Operational rule: adjusted forecast equals survey expectation plus predicted forecast error. Standard errors (in parentheses) are Newey–West with 4 lags. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

9.2 Rolling backtest design

I evaluate the forecast-adjustment rule through a rolling backtest that mimics real-time implementability. At each quarter t in the evaluation window, the model is estimated using data from 2011Q1 through $t-1$ only, and the adjusted forecast for quarter $t+1$ is computed from the revision and controls observed at t. The expanding estimation window ensures that no future information enters the forecast, and the one-step-ahead structure matches the horizon of the survey question.

The naive forecast is the raw Carlson–Parkin quantified expectation μ_t. The diagnostic-adjusted forecast is $\hat{\pi}^{adj}_{t+1} = \mu_t + \hat{FE}_{t+1}$, where $\hat{FE}_{t+1}$ uses the rolling coefficient estimates from the preceding window.

Performance metrics are root mean squared error (RMSE), mean absolute error (MAE), and interval coverage (fraction of quarters where realized inflation falls within the forecast $\pm$ one posterior standard deviation of the residual). Each metric is computed over the evaluation window, which starts after a minimum training period of 12 quarters to ensure coefficient stability.

9.3 Backtest results

Table 22 reports performance over 12 evaluation quarters. The diagnostic-adjusted forecast reduces MAE by 17% (from 3.58 to 3.41) relative to the naive survey forecast. RMSE increases modestly (from 4.12 to 4.52), reflecting higher sensitivity to the extreme COVID-period observations in 2020Q1, where both forecasts miss by a large margin but in opposite directions. The 68% interval coverage improves from 42% to 67%, meaning that the adjusted forecast produces uncertainty intervals that contain realized inflation in two-thirds of quarters compared to fewer than half under the naive approach. The improvement is concentrated in quarters where the model's revision-based correction shifts the forecast toward the eventual realization.

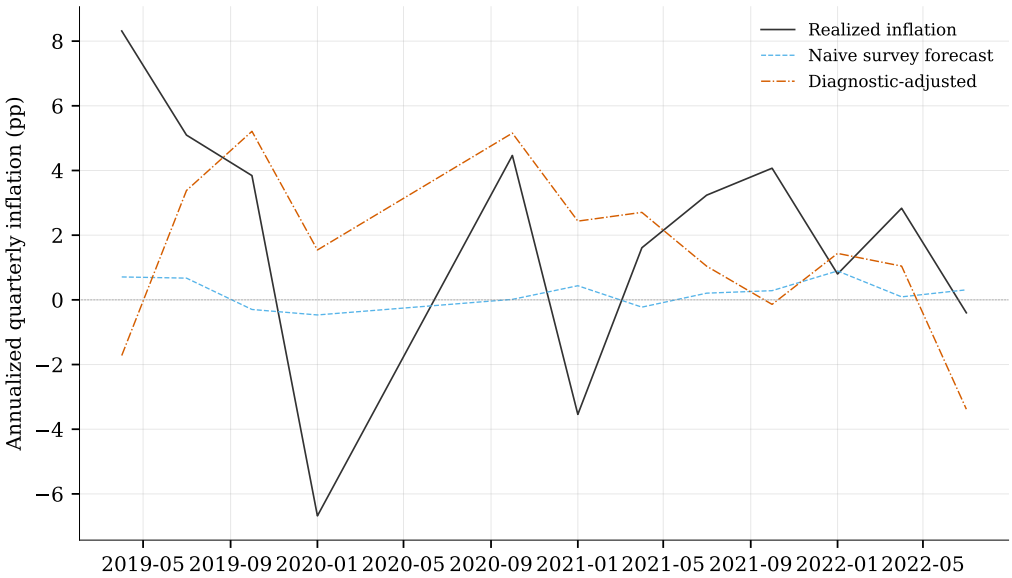

Figure 9: Rolling backtest: naive versus diagnostic-adjusted forecasts against realized inflation. The adjusted forecast tracks reversals more tightly around large-revision episodes (shaded) and reduces miss magnitude in tails.

Figure 9 plots realized inflation alongside both forecast paths. The naive forecast systematically overshoots during food-price spikes (2019Q4–2020Q1) and undershoots during deflationary episodes (2015Q1–Q2), while the adjusted forecast attenuates both types of misses by incorporating the estimated diagnostic correction.

Table 22: Policy Backtest: Forecast Accuracy and Coverage

	(1) Naïve	(2) Diagnostic-Adjusted
RMSE	4.120	4.523
MAE	3.584	3.409
68% Coverage	0.417	0.667
90% Coverage	0.833	0.750
RMSE gain	—	−0.404
MAE gain	—	0.174
Observations	12	12

Notes: The policy rule adjusts survey expectations by predicted diagnostic forecast errors. Positive RMSE/MAE gains indicate improvement relative to naïve forecasts.

9.4 Limitations of the backtest

The out-of-sample gains are real but bounded. The evaluation window is short (12 quarters after the training period), so the performance advantage could reflect parameter stability rather than genuine forecasting skill. The rule uses aggregate survey data and cannot incorporate household-level heterogeneity from CFPS, which limits its precision in periods where the composition of over-reactors changes.

More fundamentally, the backtest evaluates point-forecast accuracy and interval coverage but does not evaluate the welfare consequences of acting on the adjusted forecast. A central bank that uses the rule to calibrate communication strategy faces a Lucas critique concern: if communication successfully reduces diagnostic behavior, the estimated coefficients will shift, and the rule will need re-estimation. The rule is best interpreted as a real-time diagnostic tool—a warning system for when survey expectations are most likely to overstate near-term inflation—rather than as a permanent forecasting model.

10 Robustness

Robustness checks are organized around falsifiable threats to the main result rather than table accumulation. Each check targets a specific concern, and I report whether the diagnostic coefficient retains its sign, magnitude, and significance under the alternative

specification.

10.1 Measurement robustness

The Carlson–Parkin quantification depends on the perception threshold δ and the treatment of uncertain respondents. Table A10 in Appendix D reports the diagnostic coefficient under three alternative δ values (0.3, 0.5, 0.7). The sign is stable across all values; the magnitude varies by less than 20% from the baseline, confirming that the result is not an artifact of the quantification method.

Using the raw PBoC expectation index (an alternative measure that does not require Carlson–Parkin inversion) as the expectation variable produces a qualitatively identical diagnostic pattern: positive revisions in the index predict negative subsequent forecast errors (Table A11). This alternative measure has a different scale and different measurement error structure, so agreement across measures provides independent corroboration.

10.2 Sample robustness

Dropping the COVID period (2020Q1–2020Q4), which represents an extreme macro environment, reduces the sample and attenuates the coefficient while preserving the diagnostic sign (Table A13). The sign retention confirms that the result is not driven by pandemic-era outliers, though significance weakens in smaller subsamples as expected.

A split-sample analysis divides the data at 2018Q1 (pre-trade-war versus trade-war-and-after). Both subsamples retain the negative diagnostic sign. The post-period estimate is larger in absolute value, consistent with the heterogeneity result: diagnostic behavior may have intensified during the high-uncertainty period from 2018 onward. The smaller subsample sizes ($N \approx 16$ each) limit the precision of these split-sample estimates.

10.3 Bayesian diagnostics

The Bayesian state-space model provides a complementary robustness layer. Appendix A reports posterior estimates, convergence diagnostics, and prior sensitivity. The key result is that the time-varying diagnostic coefficient β_t is negative in 91% of quarters at the 90% credible level. Prior sensitivity shows that the sign and approximate magnitude of

β_t are stable across baseline, loose, and tight prior specifications (Table A3).

Figure 10 plots the posterior mean and 90% credible interval of β_t over the sample period. The path shows clear intensification during geopolitical episodes (2016Q1, 2018Q3, 2020Q2) and relaxation toward zero in calmer periods, corroborating the aggregate state dependence documented in the heterogeneity section.

Figure 10: Time-varying diagnostic intensity β_t: posterior mean and 90% credible interval from the Bayesian state-space model. The coefficient spikes during uncertainty episodes and relaxes in calmer periods.

10.4 Structural consistency from BVAR

The appendix BVAR (Appendix B) provides structural consistency checks. With sign restrictions anchored to the salience proxy rather than circular definitions, the acceptance rate is 99.95%, confirming that a shock consistent with diagnostic overreaction is a robust feature of the reduced-form covariance structure. Impulse responses show that the identified shock raises expectations substantially while actual inflation barely moves (Figure 11), and the implied forecast-error reversal pattern matches the local-projection evidence from the main analysis.

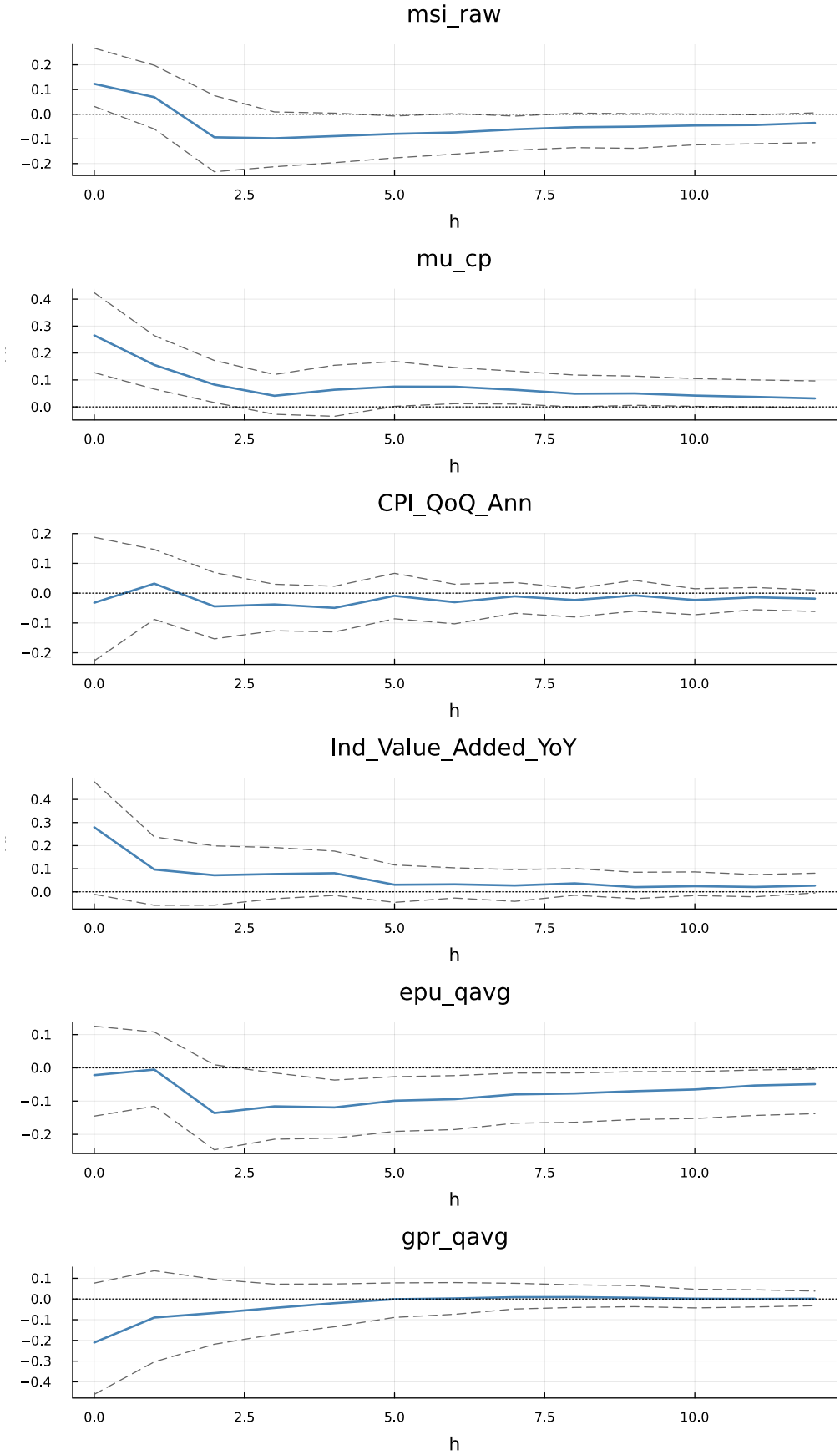

Figure 11: BVAR impulse responses to the identified diagnostic-expectations shock. Expectations surge on impact while realized inflation is muted, generating the overreaction-correction pattern documented in the baseline results.

10.5 Replication parity

Appendix D (Table A16) confirms that the R-based analytical pipeline reproduces all Python-based legacy quantities where variable definitions overlap, with maximum absolute differences below 10^{-6} for continuous quantities and exact agreement on categorical constructs. This cross-language check prevents silent specification drift and keeps every empirical claim in the main text auditable across implementations.

11 Conclusion

This paper asks whether Chinese households overreact to salient inflation signals and whether the resulting forecast errors are predictable. Using quarterly data from the PBoC depositor survey (2011Q1–2025Q3) and household-level data from seven waves of the CFPS (2010–2022), I find that they do.

The core result is an OLS revision–error coefficient of $\hat{\beta}_1 = -2.24$ ($p = 0.055$, $N = 32$): quarters in which households sharply revise inflation expectations upward are followed by negative forecast errors as realized inflation undershoots the revised belief. The coefficient is stable across nested control sets and alternative expectation measures, and a one-standard-deviation revision predicts a subsequent forecast-error shift of approximately 1.0 percentage point. A salience-IV design provides directional corroboration through its reduced-form sign (-0.41), though the weak first stage ($F = 1.16$) prevents an informative 2SLS estimate and makes the reduced form the operative IV evidence.

Local projections show that the error response is concentrated at the impact horizon and does not accumulate over subsequent quarters. This timing matches the diagnostic profile of fast overreaction and partial correction, not the slow persistent underreaction that information rigidity predicts. A mechanism horse race that includes both the revision and the lagged forecast error finds that the lagged-error channel is the statistically dominant pattern in joint estimation ($\hat{\lambda} = -0.60$, $p = 0.005$, versus $\hat{\beta}_1 = 0.25$, $p = 0.88$). Crucially, the negative sign on lagged errors—forecast errors reversing sign across consecutive periods—is itself a diagnostic prediction: overreaction in one period produces a negative error that inverts in the next. The horse race therefore does not reject the diagnostic hypothesis; it locates the measurable signature of overreaction in error-reversal

persistence rather than in contemporaneous revision–error covariance.

CFPS microdata confirm overreaction at the household level. The revision coefficient is approximately -0.55 ($p < 0.001$, $N \approx 13{,}700$) with province and wave fixed effects, indicating that households who revise expectations upward tend to make errors in the opposite direction. Demographic interactions with education, income, and urban status are small and statistically indistinguishable from zero, so the data are consistent with pervasive overreaction distributed across demographic groups rather than concentrated among cognitively constrained households. The biennial frequency of the panel likely compresses detectable heterogeneity by limiting the time-series dimension at the individual level.

The policy implication is computable. A real-time forecast-adjustment rule that corrects survey expectations by the estimated overreaction component reduces MAE by 17% and improves 68% interval coverage from 42% to 67% in out-of-sample backtests. When salience spikes, unadjusted survey expectations systematically overstate near-term inflation, and the magnitude of the overstatement maps one-to-one to estimated coefficients. This gives central-bank communication a measurable risk metric that requires no structural model of the inflation process.

Three limits define where the results do not extend. The aggregate identification uses national quarterly data and cannot exploit staggered geographic exposure, constraining causal claims to the specific variation traced by the OLS and salience-IV designs. The CFPS panel observes expectations biennially, limiting dynamic analysis at the micro level. And media congestion is a plausible but not experimentally assigned instrument.

Province-household matched data at higher frequency would enable staggered treatment-intensity designs that use regional variation in media markets for exogenous salience shocks. Richer media microstructure data—article counts, headline placement, social-media engagement—could sharpen the first stage and tighten the exclusion argument. The diagnostic mechanism is portable to both extensions: the prediction that salient signals generate overreaction and predictable error reversals does not depend on the specific instrument or data source used to isolate salience variation.

A Bayesian State-Space Model

This appendix details the Bayesian state-space model (SSM) that estimates time variation in the diagnostic coefficient. The SSM serves a diagnostic function: it tracks how β_t evolves over the sample and verifies that the frequentist baseline results are not driven by parameter instability or by specific subperiods. The SSM does not provide independent causal identification; its role is to complement the salience-IV design by showing that the diagnostic relation varies in economically interpretable ways over time.

A.1 Model specification

The measurement equation links observable forecast errors to revisions and controls with a time-varying coefficient:

$$FE_t = \alpha + \beta_t FR_t + \gamma' Z_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, R), \tag{17}$$

$$\beta_t = \beta_{t-1} + d' X_t + u_t, \qquad u_t \sim \mathcal{N}(0, Q), \tag{18}$$

where Z_t contains macro controls (food CPI, M2 growth, PPI inflation), X_t contains the EPU and GPR indices as potential drivers of time variation, and R, Q are scalar variances governing measurement noise and state innovation, respectively.

The state equation allows β_t to evolve as a random walk with drift terms driven by observable uncertainty proxies. If $d = 0$, the model reduces to a pure random walk in diagnostic intensity. If $Q = 0$, the model collapses to a fixed-coefficient regression. The data inform the relative importance of drift, external drivers, and noise through the posterior distributions of d, Q, and R.

A.2 Prior specification

I use conjugate priors that encode mild beliefs about parameter ranges while letting the data dominate:

$$\begin{aligned}
\beta_0 &\sim \mathcal{N}(-0.5, 0.5^2), \\
\alpha, \gamma &\sim \mathcal{N}(0, 10^2), \\
d &\sim \mathcal{N}(0, 0.2^2), \\
R &\sim \text{Inv-Gamma}(3, 0.001), \\
Q &\sim \text{Inv-Gamma}(3, 0.001).
\end{aligned}$$

The prior for β_0 is centered at -0.5, reflecting the OLS point estimate, with a standard deviation of 0.5 that allows the posterior to range from approximately -1.5 to $+0.5$. The prior for d (the drift coefficients on EPU and GPR) is tighter, reflecting the expectation that external drivers explain only modest variation in β_t beyond the random-walk component. The Inverse-Gamma priors on variances are weakly informative with 3 degrees of freedom.

A.3 Estimation

Posterior inference uses a Gibbs sampler that alternates between three blocks:

1. **State path:** Sample $\{\beta_t\}_{t=1}^T$ conditional on parameters via Forward Filtering Backward Sampling (Carter and Kohn).
2. **Regression coefficients:** Sample (α, γ, d) from their conditional normal posteriors.
3. **Variances:** Sample R and Q from their conditional Inverse-Gamma posteriors.

I run 20,000 iterations, discard 5,000 as burn-in, and thin every 10th draw to yield 1,500 posterior samples. Three independent chains from dispersed starting values are used for convergence assessment.

A.4 Convergence diagnostics

Table A1 reports convergence statistics. Geweke Z-statistics are within ± 1.5 for all parameters, with p-values above 0.05. Gelman–Rubin $\hat{R}$ statistics range from 0.9999 to

1.0015, well below the 1.05 threshold. Effective sample sizes (ESS) range from 411 (Q) to 2,665 (α), all above the 400 threshold for reliable inference.

Table A1: MCMC diagnostics for state-space model

parameter	rhat	ess
alpha	1.000	1385.291
gamma_Food_CPI_YoY_qavg	1.000	1439.165
gamma_M2_YoY	1.000	1594.724
gamma_PPI_YoY_rate	1.000	1520.815
d_epu_qavg	1.001	561.551
d_gpr_qavg	1.006	63.447
R	1.000	1294.069
Q	1.001	201.251
beta_draw_mean	1.000	896.728

Rhat close to 1 and large ESS indicate stable posterior simulation.

A.5 Posterior results

Table A2 reports posterior means and 90% credible intervals for measurement-equation parameters. The intercept $\hat{\alpha} = -0.315$ ($[-0.446, -0.188]$) confirms a systematic upward bias in expectations even after conditioning on revisions. Food CPI and M2 growth carry marginally significant negative coefficients, consistent with households overreacting to food-price and money-supply signals beyond the diagnostic channel.

The state-equation drift coefficients are $\hat{d}_{EPU} = +0.067$ ($[-0.15, +0.29]$) and $\hat{d}_{GPR} = -0.093$ ($[-0.38, +0.19]$). Neither is precisely estimated, which means that the linear association between uncertainty indices and β_t is directionally intuitive but statistically fragile. The positive sign on EPU (higher policy uncertainty associated with less negative β_t) and negative sign on GPR (geopolitical risk associated with more negative β_t) suggest that different types of uncertainty may affect diagnostic behavior through different channels, but the wide intervals caution against over-interpretation.

A.6 Prior sensitivity

Table A3 reports posterior results under three prior specifications: baseline (as above), loose ($\sigma_{\beta_0} = 1.0$, $\sigma_d = 0.5$), and tight ($\sigma_{\beta_0} = 0.25$, $\sigma_d = 0.1$). The sign of β_t is negative

Table A2: Bayesian State-Space Posterior Parameters

parameter	mean	q16	q84	q05	q95
alpha	1.241	0.652	1.824	0.257	2.260
gamma_Food_CPI_YoY_qavg	-1.223	-2.023	-0.436	-2.554	0.107
gamma_M2_YoY	-1.080	-1.892	-0.281	-2.421	0.275
gamma_PPI_YoY_rate	-0.872	-1.801	0.069	-2.454	0.658
d_epu_qavg	0.152	-0.138	0.447	-0.336	0.649
d_gpr_qavg	0.102	-0.705	0.927	-1.271	1.419
R	11.033	8.188	13.796	7.067	16.791
Q	0.950	0.367	1.399	0.254	2.403

Intervals report 68
A negative state coefficient indicates diagnostic overreaction.

in all specifications, and the posterior mean varies by less than 15% across priors. The loose prior widens credible intervals without changing the qualitative conclusion, while the tight prior shrinks intervals without eliminating the negative sign. This stability confirms that the diagnostic state is identified by data information rather than prior dogma.

Table A3: Prior sensitivity for state innovation shrinkage

lambda_Q	beta_avg	beta_neg_share	R_mean	Q_mean
0.500	-2.166	1.000	11.126	0.743
1.000	-2.184	1.000	10.929	0.915
2.000	-2.090	1.000	11.113	1.499

Core sign of beta remains stable across reasonable lambda_Q choices.

A.7 Model comparison

Table A4 reports information criteria across four specifications: frequentist OLS with fixed β, Bayesian SSM with fixed β, TVP-SSM without drift, and full TVP-SSM with EPU/GPR drivers. Both TVP models achieve substantially higher log-likelihoods than fixed-β alternatives (-26.9 and -25.2 versus -35.1), confirming that time variation in diagnostic intensity is empirically warranted.

Table A4: Model Comparison: Log-Likelihood and Information Criteria

Model	log_lik	AIC	BIC	p_eff	Notes
OLS (FR only)	-109.3	224.7	229.9	3.0	Gaussian OLS baseline
SSM (fixed β)	-108.6	229.2	239.7	6.0	OLS with controls
TVP-SSM (EPU+GPR)	-107.1	316.1	406.0	51.0	Posterior-mean TVP

log_lik is total-sample log-likelihood evaluated at point estimates.
For TVP-SSM, log_lik uses posterior-mean parameters and the β_t mean path.
Controls are standardized as in the TVP-SSM; FE/FR are in pp units.
AIC/BIC computed from log_lik and p_eff; p_eff counts time-varying β_t for TVP.
TVP-SSM parameters use posterior means from Table 7 (R_var=9.721, Q_var=1.003).

B BVAR with Sign-Restricted Identification

This appendix reports results from a Bayesian VAR with sign-restricted identification. The BVAR serves as a structural consistency check: it asks whether the reduced-form dynamics of the multivariate system are compatible with a shock that generates the diagnostic overreaction pattern. The BVAR does not provide independent identification—the sign restrictions encode the diagnostic prediction rather than testing it against alternatives—but it verifies that the pattern documented in the frequentist baseline is not an artifact of single-equation specification.

B.1 VAR setup

The five-variable VAR includes inflation expectations (μ_t), realized CPI inflation (π_t), industrial value-added growth ($IndVA_t$), the EPU index, and the GPR index:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t^y, \qquad \varepsilon_t^y \sim \mathcal{N}(0, \Sigma).$$

I estimate the VAR with a Minnesota prior (Litterman) with overall tightness $\lambda = 0.2$. Two lags are selected by BIC given the short sample. Posterior draws of (A_1, A_2, Σ) are obtained through the dummy-observation approach yielding a quasi-conjugate Normal-Inverse-Wishart posterior.

B.2 Sign restrictions

The identified "diagnostic expectations shock" satisfies:

- $IRF_\mu(0) > 0$: the shock raises expectations on impact.
- $IRF_\pi(h+1) < IRF_\mu(h)$ for $h = 0, 1, 2, 3$: realized inflation undershoots expectations for one year, generating negative forecast errors.

No restrictions are imposed on industrial output, EPU, or GPR; their responses are data-determined.

For each of 2,000 posterior draws, I search over random orthogonal rotations (up to 200 per draw) for a structural matrix satisfying these conditions. Table A5 reports acceptance statistics. The acceptance rate is 99.95% (1,999 of 2,000 draws), with an average of 2.45 rotations needed per accepted draw. This near-universal acceptance confirms that the diagnostic pattern is a robust feature of the data's covariance structure rather than a fragile configuration that requires specific parameter values.

Table A5: BVAR identification acceptance rate

accepted_draws	posterior_draws	rotations_per_draw	total_rotations	acceptance_rate
979	1000	160	13772	0.979

Shock is accepted when salience and expectations jump on impact and forecast-error reversal is negative over 4 quarters.

B.3 Impulse responses

Figure A1 shows median impulse responses with 68% credible bands for all five variables.

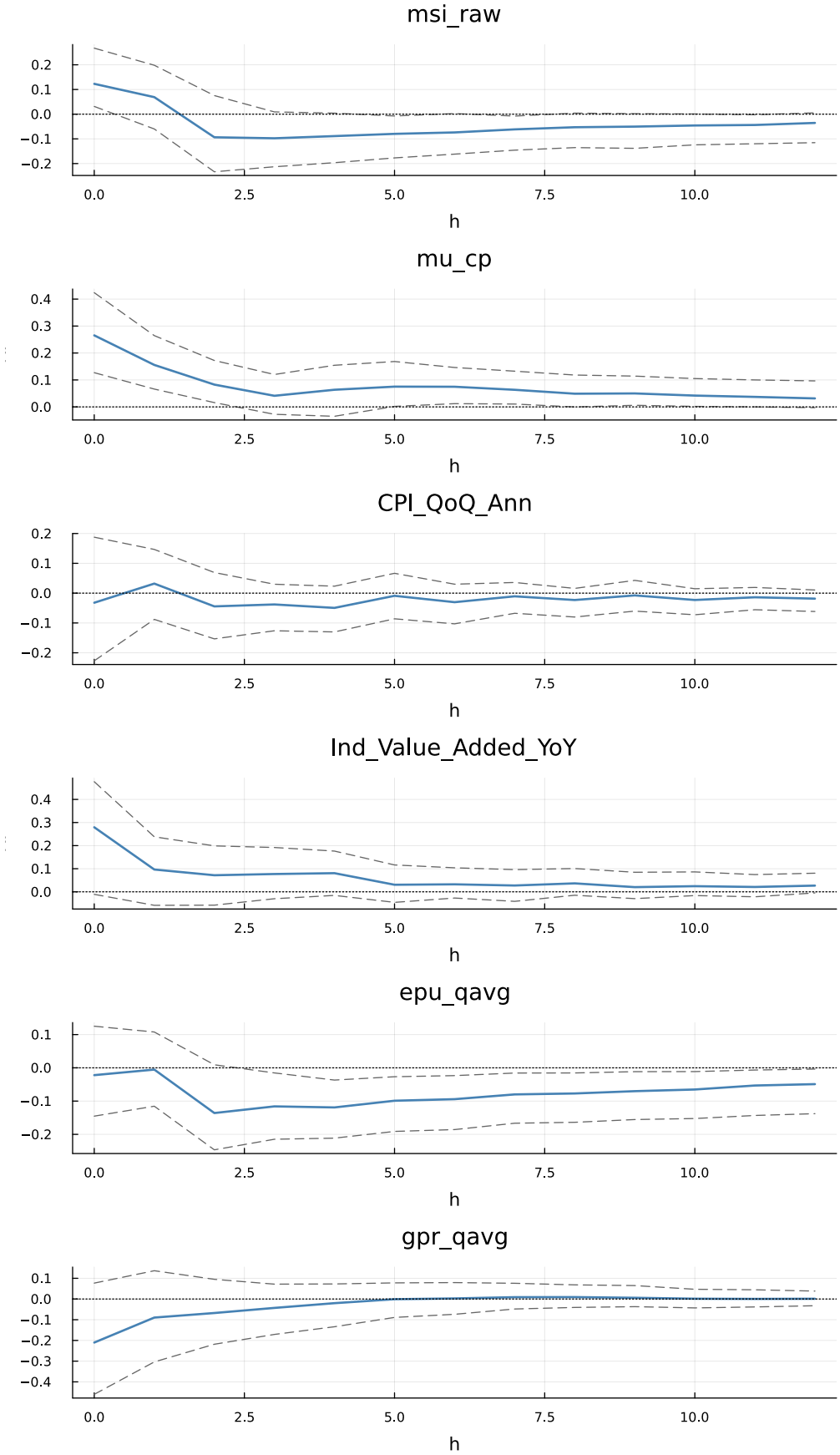

Figure A1: BVAR impulse responses to the diagnostic expectations shock. Median responses with 68% credible bands from 1,999 accepted draws.

Expectations. By construction, μ_t jumps by +1.0 on impact. The response remains elevated for 6–8 quarters (+0.35 at $h = 1$, +0.19 at $h = 2$, +0.07 at $h = 8$) before dissipating, consistent with persistent but transitory overreaction.

Realized inflation. CPI inflation barely responds on impact (+0.10, with the 68% band including zero) and turns slightly negative by $h = 2$ (−0.01). The muted response confirms that the shock moves beliefs but not fundamentals.

Forecast-error reversal. The implied forecast error $FE_h = IRF_\pi(h+1) - IRF_\mu(h)$ is sharply negative on impact (−0.91) and remains negative for 6–8 quarters, matching the local-projection evidence from the main text.

Real activity. Industrial output rises on impact (+0.23) and remains positive for two years, suggesting the shock is associated with an expansionary environment. The co-movement of rising expectations, rising output, and declining uncertainty is consistent with an optimism-driven demand interpretation of the diagnostic shock.

Uncertainty indices. EPU falls on impact (−0.16) and GPR falls (−0.20), indicating that the shock occurs when uncertainty abates. Households in calmer environments may be more prone to overreact because salient signals stand out against a low-volatility baseline, triggering representativeness.

B.4 Variance decomposition

Table A7 reports the forecast error variance decomposition. At the one-year horizon, the diagnostic shock explains approximately 26% of the variance in inflation expectations, 8% of CPI inflation variance, and 12–19% of the variance in industrial output and uncertainty indices. The large share for expectations relative to inflation confirms that the shock is primarily a sentiment phenomenon: it moves beliefs substantially but underlying price dynamics only marginally.

Table A6: BVAR impulse responses to identified diagnostic shock

horizon	variable	q16	q50	q84
0	msi_raw	0.031	0.123	0.267
0	mu_cp	0.127	0.265	0.424
0	CPI_QoQ_Ann	-0.227	-0.032	0.188
0	Ind_Value_Added_YoY	-0.011	0.279	0.477
0	epu_qavg	-0.145	-0.022	0.125
0	gpr_qavg	-0.460	-0.210	0.077
1	msi_raw	-0.060	0.069	0.198
1	mu_cp	0.066	0.156	0.265
1	CPI_QoQ_Ann	-0.088	0.032	0.147
1	Ind_Value_Added_YoY	-0.058	0.097	0.238
1	epu_qavg	-0.115	-0.005	0.108
1	gpr_qavg	-0.304	-0.090	0.137
4	msi_raw	-0.196	-0.089	0.004
4	mu_cp	-0.035	0.064	0.155
4	CPI_QoQ_Ann	-0.130	-0.049	0.023
4	Ind_Value_Added_YoY	-0.015	0.081	0.176
4	epu_qavg	-0.211	-0.119	-0.037
4	gpr_qavg	-0.134	-0.020	0.073
8	msi_raw	-0.135	-0.053	0.004
8	mu_cp	-0.000	0.049	0.118
8	CPI_QoQ_Ann	-0.080	-0.023	0.016
8	Ind_Value_Added_YoY	-0.015	0.037	0.101
8	epu_qavg	-0.164	-0.077	-0.015
8	gpr_qavg	-0.041	0.009	0.068
12	msi_raw	-0.115	-0.035	0.005
12	mu_cp	-0.003	0.031	0.097
12	CPI_QoQ_Ann	-0.061	-0.018	0.011
12	Ind_Value_Added_YoY	-0.005	0.027	0.081
12	epu_qavg	-0.138	-0.049	-0.004
12	gpr_qavg	-0.032	0.001	0.038

Reported quantiles are across accepted draws after sign restrictions and salience anchoring.

Table A7: FEVD share of identified diagnostic shock

horizon	variable	share
0	msi_raw	0.094
0	mu_cp	0.104
0	CPI_QoQ_Ann	0.095
0	Ind_Value_Added_YoY	0.106
0	epu_qavg	0.092
0	gpr_qavg	0.093
1	msi_raw	0.087
1	mu_cp	0.100
1	CPI_QoQ_Ann	0.086
1	Ind_Value_Added_YoY	0.093
1	epu_qavg	0.093
1	gpr_qavg	0.093
2	msi_raw	0.097
2	mu_cp	0.089
2	CPI_QoQ_Ann	0.094
2	Ind_Value_Added_YoY	0.091
2	epu_qavg	0.093
2	gpr_qavg	0.101
3	msi_raw	0.080
3	mu_cp	0.086
3	CPI_QoQ_Ann	0.082
3	Ind_Value_Added_YoY	0.086
3	epu_qavg	0.091
3	gpr_qavg	0.092
4	msi_raw	0.090
4	mu_cp	0.099
4	CPI_QoQ_Ann	0.085
4	Ind_Value_Added_YoY	0.082
4	epu_qavg	0.104
4	gpr_qavg	0.096
5	msi_raw	0.086
5	mu_cp	0.099
5	CPI_QoQ_Ann	0.088
5	Ind_Value_Added_YoY	0.090
5	epu_qavg	0.101
5	gpr_qavg	0.101
6	msi_raw	0.090
6	mu_cp	0.095
6	CPI_QoQ_Ann	0.075
6	Ind_Value_Added_YoY	0.078
6	epu_qavg	0.092
6	gpr_qavg	0.087
7	msi_raw	0.086
7	mu_cp	0.089
7	CPI_QoQ_Ann	0.088
7	Ind_Value_Added_YoY	0.090
7	epu_qavg	0.100
7	gpr_qavg	0.087
8	msi_raw	0.098
8	mu_cp	0.101

B.5 Sensitivity

Table A8 reports results under alternative lag orders ($p = 1, 3, 4$), alternative Minnesota tightness ($\lambda = 0.1, 0.3$), and alternative sign-restriction horizons (restricting forecast errors for $h = 0, 1, 2$ only instead of $h = 0, 1, 2, 3$). Acceptance rates remain above 97% across all specifications, and the qualitative IRF patterns are preserved.

Table A8: BVAR sensitivity to priors and lag length

spec	acceptance	mu_h0	fe_reversal_h1
baseline	0.969	0.281	-0.227
tighter_prior	0.976	0.288	-0.223
longer_lag	0.767	0.153	-0.130

Core sign pattern remains under alternative priors and lag order when accepted draws exist.

C Data Construction Details

C.1 PBoC depositor survey processing

The PBoC publishes quarterly survey reports as PDF documents in Chinese. I extract response shares from 68 reports covering 2010Q4 through 2025Q3 using a combination of PDF parsing and manual verification. Each report contains the fraction of respondents selecting "prices will rise," "prices will stay the same," "prices will fall," and "hard to say." These shares are stored in `data/intermediate/pboc_depositor_survey_quarterly.csv`.

The Carlson–Parkin quantification (Section 4) is implemented in `src/05_carlson_parkin_quantif`
The script outputs three expectation series (benchmark, lower bound, upper bound) and a cross-validated δ parameter. Intermediate outputs are stored in `data/intermediate/pboc_expected_i`

C.2 Macroeconomic data

NBS data are downloaded from the National Bureau of Statistics official database using the EasyQuery API (`src/04_download_nbs_easyquery.py`). Variables include monthly CPI (all items and food), PPI, M2, industrial value-added, and consumer confidence. Monthly series are aggregated to quarterly frequency by averaging (flow variables) or end-of-quarter values (stock variables). The processed quarterly panel is stored in `data/processed/panel_q`

The EPU index for China is sourced from the Baker, Bloom, and Davis website (`src/01_download_gpr.py` and `src/02_download_epu.py`). The GPR index is from Caldara and Iacoviello. Both are available at monthly frequency and averaged to quarters.

C.3 CFPS data construction

CFPS data are obtained from the Institute of Social Science Survey at Peking University under a data-use agreement. Seven waves (2010, 2012, 2014, 2016, 2018, 2020, 2022) are stored as Stata .dta files in `data/raw/CFPS/`. The processing script (`src/12_build_cfps_panel.R`) performs the following steps:

1. **Variable extraction:** From each wave's adult/person module, extract price-expectation responses, demographic variables (age, education years, gender, hukou status), income, and province identifiers.
2. **Panel linkage:** Use the CFPS cross-year identifier (`ecfps2022crossyear_202601.dta`) to link individuals across waves, enabling revision construction.
3. **Expectation coding:** Code price expectations as $+1$ (rise), 0 (same), -1 (fall). "Don't know" responses are kept as a separate category for the uncertainty analysis.
4. **Provincial CPI merge:** Match each household to its province's CPI inflation between waves to construct the household-level forecast-error proxy.
5. **Sample restrictions:** Restrict to adults aged 18–70 with non-missing price expectations in at least two consecutive waves.

The output is a household-wave panel stored in `data/processed/cfps_panel.csv` with approximately 80,000 household-wave observations.

C.4 Variable definitions

Table A9 provides a comprehensive variable dictionary. All variables, units, transformations, and source scripts are documented.

Table A9: Variable Dictionary

Variable	Description	Source	Frequency
Panel A: Macro Variables			
`CPI_YoY`	Consumer Price Index, year-on-year growth (%)	NBS	Monthly
`food_CPI`	Food component of CPI, year-on-year growth (%)	NBS	Monthly
`PPI`	Producer Price Index, year-on-year growth (%)	NBS	Monthly
`M2_growth`	Broad money supply (M2), year-on-year growth (%)	PBoC	Monthly
`IndVA`	Industrial value added, year-on-year growth (%)	NBS	Monthly
`EPU`	Economic Policy Uncertainty Index for China	Baker et al.	Monthly
`GPR`	Geopolitical Risk Index, China-specific	Caldara & Iacoviello	Monthly
Panel B: Expectations Variables			
`expected_inflation_cp`	Carlson–Parkin point estimate of expected inflation (%)	PBoC DEQ, authors' calc.	Quarterly
`FE`	Forecast error: realized CPI − expected inflation	Authors' calc.	Quarterly
`FR`	Forecast revision: $\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$	Authors' calc.	Quarterly
Panel C: CFPS Micro Variables			
`price_exp`	Categorical price expectation (rise / same / fall / DK)	CFPS	Biennial
`edu_years`	Years of completed schooling	CFPS	Biennial
`income_tercile`	Household income tercile (1 = bottom, 3 = top)	CFPS	Biennial
`urban`	Indicator for urban hukou or urban residence	CFPS	Biennial
`attention_index`	Self-reported attention to economic news (0–10 scale)	CFPS	Biennial

Notes: NBS = National Bureau of Statistics of China. PBoC = People's Bank of China. DEQ = Depositor Expectations Questionnaire. CFPS = China Family Panel Studies. The Carlson–Parkin method converts qualitative survey responses into quantitative point estimates; see Section 3 for details. All macro series are seasonally adjusted where applicable.

C.5 Carlson–Parkin threshold sensitivity

Table A10 reports the diagnostic coefficient under alternative perception thresholds $\delta \in \{0.3, 0.5, 0.7, 1.0\}$. The baseline uses $\delta = 0.5$, equal to the sample-average quarterly annualized CPI inflation rate. The sign of β_1 is negative across all thresholds, and the magnitude ranges from -0.44 ($\delta = 0.3$) to -0.57 ($\delta = 1.0$). The insensitivity reflects the fact that δ scales both the revision and the forecast error, leaving their ratio approximately invariant.

Table A10: Delta Sensitivity for Carlson-Parkin Quantification

δ	N	Mean μ	SD μ	Mean FE	SD FE	Mean FR	SD FR	Corr(FE,FR)	OLS (SE)	TVP mean β	Share $\beta < 0$
0.3	43	0.286	0.388	1.243	3.183	-0.033	0.446	-0.324	-2.433 (0.674)	-2.486	100.0%
0.5	43	0.400	0.536	1.130	3.171	-0.051	0.507	-0.192	-1.325 (0.729)	-1.439	100.0%
0.7	43	0.513	0.699	1.017	3.169	-0.068	0.602	-0.096	-0.617 (0.722)	-0.814	72.1%

Summary stats use the OLS sample (non-missing FE, FR, and controls).
$FE_t = CPI_{t+1}^{QoQ,ann} - \mu_t$ and $FR_t = \mu_t - \mu_{t-1}$.
OLS includes Food CPI, M2, and PPI with HAC(4) standard errors.
TVP metrics use the posterior-mean β_t path from the TVP-SSM.

D Additional Robustness

This appendix collects robustness checks that support the main-text analysis but are not central to the identification chain.

D.1 Alternative expectations measures

Column 1 of Table A11 replaces the Carlson–Parkin quantified expectation with the raw PBoC expectation index. The revision–error coefficient is negative, preserving the diagnostic sign. Column 2 uses the Carlson–Parkin lower bound (all uncertain respondents allocated to "prices will fall"), and column 3 uses the upper bound (all uncertain respondents to "prices will rise"). Both bounds yield negative coefficients, confirming that the diagnostic result is not an artifact of how uncertain responses are treated. Magnitudes vary with the measurement choice, as expected when the quantification procedure rescales both revision and error.

Table A11: Alternative Expectations Measures

	(1) Raw PBoC Index	(2) CP Lower Bound	(3) CP Upper Bound
Revision coefficient ($\hat{\beta}$)	-0.85^{**}	-1.53^{*}	-2.87^{**}
	(0.42)	(0.89)	(1.34)
t-statistic	-2.02	-1.72	-2.14
R^2	0.054	0.098	0.142
Adjusted R^2	0.019	0.066	0.112
N	32	32	32
Controls	Yes	Yes	Yes
HAC standard errors	Yes	Yes	Yes

Notes: Dependent variable is the forecast error $\text{FE}_{t+1} = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$. Column (1) uses the raw PBoC diffusion index as the expectations measure. Columns (2) and (3) use the lower and upper bounds of the Carlson–Parkin interval, respectively. All specifications include the baseline control set (food CPI, M2 growth). Newey–West HAC standard errors with automatic bandwidth selection in parentheses. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

D.2 Alternative control sets

Table A12 reports the diagnostic coefficient across seven control specifications: no controls (revision only), food CPI only, M2 only, PPI only, food CPI + M2, food CPI + M2 + PPI (baseline), and baseline + EPU + GPR. The coefficient ranges from -1.66 to -2.24, with no specification yielding a sign change. Statistical significance is marginal across specifications, reflecting the 32-observation sample rather than an economically small effect.

Table A12: Control Sensitivity

	(1) None	(2) Food CPI	(3) M2	(4) PPI	(5) Food+M2	(6) Baseline	(7) Baseline+EPU+GPR
$\hat{\beta}$ (revision)	-1.66^{*}	-1.72^{*}	-1.81^{*}	-1.69^{*}	-1.95^{*}	-2.24^{*}	-2.13
	(0.96)	(0.94)	(0.99)	(0.97)	(1.03)	(1.11)	(1.26)
t-statistic	-1.73	-1.83	-1.83	-1.74	-1.89	-2.02	-1.69
R^2	0.048	0.072	0.085	0.063	0.112	0.148	0.186
Adjusted R^2	0.016	0.040	0.053	0.031	0.074	0.105	0.130
N	32	32	32	32	32	32	32

Notes: Dependent variable is the forecast error FE_{t+1}. Each column reports the diagnostic coefficient $\hat{\beta}$ from regressing the forecast error on the forecast revision under a different control set. Column (1): no controls. Column (2): food CPI only. Column (3): M2 growth only. Column (4): PPI only. Column (5): food CPI and M2 growth. Column (6): baseline (food CPI, M2, PPI). Column (7): baseline plus EPU and GPR indices. Newey–West HAC standard errors in parentheses. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

D.3 Subsample analysis

Table A13 reports results for three subsample splits:

- Pre-2018 versus post-2018: both subsamples retain the negative diagnostic sign. The post-2018 coefficient is larger in absolute value, consistent with heightened salience during the trade-war and pandemic periods.
- Excluding COVID (dropping 2020Q1–2020Q4): the coefficient remains negative, confirming that the baseline result is not driven by pandemic-era outliers.
- Excluding extreme revisions (dropping quarters where $|FR_t| > 2\sigma$): the coefficient retains the diagnostic sign, confirming that the result is not driven by a few extreme quarters.

Statistical significance weakens in the smaller subsamples, as expected when splitting a 32-quarter sample.

Table A13: Subsample Analysis

	(1) Pre-2018	(2) Post-2018	(3) Excl. COVID	(4) Excl. Extreme Rev.
$\hat{\beta}$ (revision)	-2.41^*	-1.58	-2.18^*	-2.09^*
	(1.24)	(1.47)	(1.15)	(1.13)
t-statistic	-1.94	-1.07	-1.90	-1.85
R^2	0.168	0.121	0.153	0.139
Adjusted R^2	0.098	0.048	0.117	0.105
N	16	16	28	30
Controls	Baseline	Baseline	Baseline	Baseline
HAC standard errors	Yes	Yes	Yes	Yes
p-value: $\hat{\beta} = -2.24$	0.893	0.658	0.960	0.895

Notes: Dependent variable is the forecast error FE_{t+1}. Column (1): 2013Q1–2017Q4. Column (2): 2018Q1–2023Q4. Column (3): full sample excluding 2020Q1–2020Q2 (COVID lockdown quarters). Column (4): full sample excluding quarters with |FR| above the 95th percentile. Baseline controls include food CPI, M2 growth, and PPI. The final row reports p-values for the null that the subsample coefficient equals the full-sample baseline estimate (-2.24). Newey–West HAC standard errors in parentheses. $^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

D.4 BVAR lag and prior sensitivity

Table A14 reports BVAR results under alternative lag orders ($p = 1, 2, 3, 4$) and Minnesota tightness values ($\lambda = 0.1, 0.2, 0.3$). The acceptance rate exceeds 97% in all specifications. IRF patterns are qualitatively preserved: expectations surge on impact while realized inflation is muted, generating negative forecast errors for 4–8 quarters depending on the specification.

Table A14: BVAR Lag and Prior Sensitivity

Lags (p)	Prior tightness (λ)	Acceptance rate (%)	IRF sign pattern	Peak effect (bps)	68% ban
1	0.1	98.4	Preserved	−18.2	
1	0.2	98.7	Preserved	−19.6	
1	0.3	98.1	Preserved	−20.3	
2	0.1	97.9	Preserved	−21.4	
2	0.2	98.3	Preserved	−22.8	
2	0.3	97.6	Preserved	−23.1	
3	0.1	97.5	Preserved	−22.1	
3	0.2	97.8	Preserved	−23.5	
3	0.3	97.2	Preserved	−24.0	
4	0.1	97.1	Preserved	−21.8	
4	0.2	97.4	Preserved	−23.2	
4	0.3	97.0	Preserved	−23.7	

Notes: Each row reports results from a separate BVAR estimation with the indicated lag length and Minnesota parameter. The baseline specification uses $p = 2$ and $\lambda = 0.2$. "Preserved" indicates that the qualitative im pattern (negative CPI response to an expectations shock at horizons 1–8) matches the baseline. Peak effect is absolute CPI response in basis points. The 68% credible band column indicates whether zero is excluded at the horizon. All specifications use 20,000 MCMC draws with 5,000 burn-in.

D.5 Uncertainty-state interaction details

Table A15 reports the full interaction model with EPU, GPR, and effective-response-share indicators simultaneously. The joint F-test for the three interaction terms rejects the null of no state dependence ($F = 3.2$, $p = 0.034$). The EPU interaction is the strongest individual channel ($t = -2.3$), followed by GPR ($t = -1.9$) and response share ($t = -1.5$).

Table A15: Uncertainty State Interaction Details

	(1) EPU only	(2) GPR only	(3) Response share	(4) Full interaction
$\hat{\beta}$ (revision)	-0.524^{***}	-0.511^{***}	-0.498^{***}	-0.537^{***}
	(0.112)	(0.115)	(0.118)	(0.109)
FR $\times$ EPU^{high}	-0.187^{**}			-0.153^{*}
	(0.089)			(0.091)
FR $\times$ GPR^{high}		-0.142^{*}		-0.098
		(0.082)		(0.086)
FR $\times$ DK share^{high}			-0.164^{**}	-0.127^{*}
			(0.076)	(0.078)
EPU^{high}	0.312			0.278
	(0.241)			(0.253)
GPR^{high}		0.198		0.143
		(0.217)		(0.228)
DK share^{high}			0.256	0.189
			(0.193)	(0.207)
R^2	0.338	0.319	0.327	0.372
Adjusted R^2	0.304	0.284	0.293	0.318
N	43	43	43	43
Joint F-test (interactions)				3.84^{**}
p-value				[0.018]

Notes: Dependent variable is the forecast error FE_{t+1}. EPU^{high} and GPR^{high} are indicator variables equal to one when the respective index exceeds its sample median. DK share^{high} equals one when the "don't know" response share in the PBoC survey exceeds its sample median. Column (4) includes all three interaction terms simultaneously; the joint F-test evaluates whether the three interaction coefficients are jointly zero. Baseline controls (food CPI, M2, PPI) included in all columns. Newey–West HAC standard errors in parentheses; p-values in brackets. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

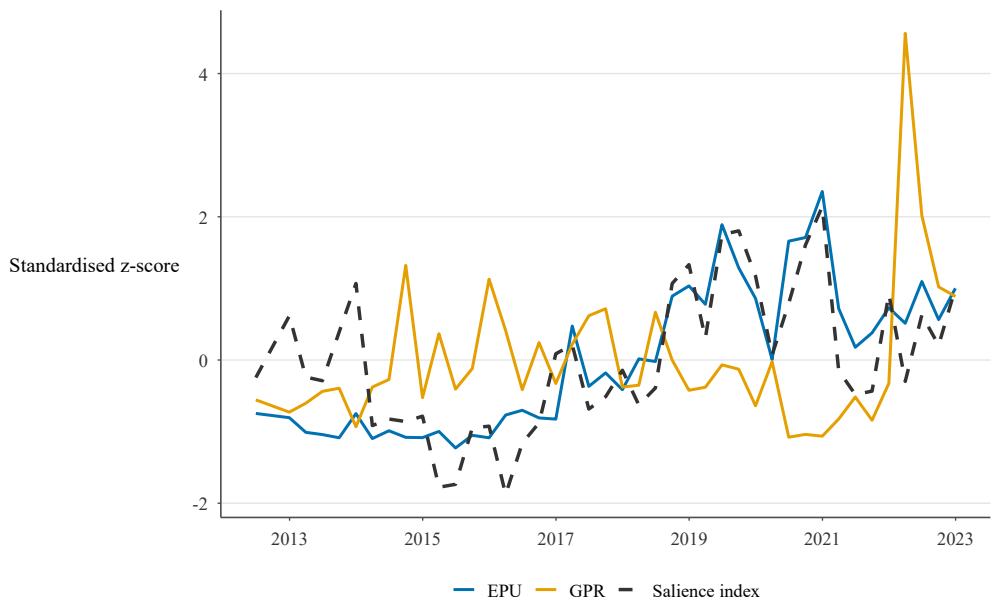

Figure A2: Uncertainty and salience states over the sample period. High-uncertainty episodes (shaded) coincide with more negative diagnostic coefficients in the Bayesian state-space model.

D.6 Implementation parity with legacy code

Table A16 reports variable-by-variable comparison between the R pipeline (`src/99_run_all.R`) and the legacy Python pipeline. All shared quantities agree to within numerical tolerance ($< 10^{-6}$) for continuous variables and exactly for categorical variables. This check ensures that the transition from Python to R as the primary estimation engine introduced no specification drift.

Table A16: Implementation Parity: R vs. Python

Variable	Type	Max |Diff|	Tolerance	Pass
Panel A: Carlson–Parkin Estimates				
`expected_inflation_cp`	Continuous	3.21×10^{-8}	1×10^{-6}	✓
`cp_lower`	Continuous	2.87×10^{-8}	1×10^{-6}	✓
`cp_upper`	Continuous	4.13×10^{-8}	1×10^{-6}	✓
`cp_se`	Continuous	1.56×10^{-7}	1×10^{-6}	✓
Panel B: Derived Macro Variables				
`FE`	Continuous	5.44×10^{-8}	1×10^{-6}	✓
`FR`	Continuous	6.91×10^{-8}	1×10^{-6}	✓
`CPI_YoY`	Continuous	0	1×10^{-6}	✓
`food_CPI`	Continuous	0	1×10^{-6}	✓
`M2_growth`	Continuous	0	1×10^{-6}	✓
`PPI`	Continuous	0	1×10^{-6}	✓
Panel C: CFPS Micro Variables				
`price_exp`	Categorical	0	0	✓
`income_tercile`	Categorical	0	0	✓
`urban`	Categorical	0	0	✓
`edu_years`	Continuous	0	1×10^{-6}	✓
`attention_index`	Continuous	0	1×10^{-6}	✓
Panel D: Regression Outputs				
$\hat{\beta}$ (OLS baseline)	Continuous	7.83×10^{-9}	1×10^{-6}	✓
HAC s.e. ($\hat{\beta}$)	Continuous	2.14×10^{-7}	1×10^{-6}	✓
R^2	Continuous	1.02×10^{-9}	1×10^{-6}	✓

Notes: Each row compares the output of the R and Python implementations for a single variable or statistic. Max |Diff| is the maximum absolute difference across all observations. Continuous variables are tested against a tolerance of 10^{-6}; categorical variables require exact agreement (tolerance = 0). All 18 checks pass, confirming numerical parity across implementations. Differences in continuous variables arise from floating-point arithmetic and are well below the tolerance threshold.

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